A JAX–PETSc Toolset for Nonlinear Finite-Element Energy Solves: Derivative Routes, Sparse Assembly, and Solver Policy Across Mechanics, Scalar PDEs, and Topology Optimization

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Abstract

Large nonlinear finite-element energy problems couple several numerical difficulties: local energies and constitutive laws need reliable derivatives, Newton-type methods need robust globalization, and distributed runs need sparse assembly, Krylov solvers, and preconditioners that match the mechanics of the problem. We present a computational toolset for studying these choices together in nonlinear finite-element solves and energy-based topology optimization. The mainline realization is a JAX+PETSc stack in which local energies and constitutive laws are written in JAX, while sparse assembly, nonlinear globalization, Krylov linear solves, and preconditioning are handled by PETSc. The same benchmark suite also includes pure JAX references and FEniCS references where they are available, so the distributed stack can be compared against compact differentiable and high-level FEM formulations. The paper studies nonlinear scalar problems (p-Laplace and Ginzburg–Landau), finite-strain hyperelasticity, two- and three-dimensional Mohr–Coulomb plasticity, and topology optimization. Across these families, the methodological comparison centers on three routes to second-order information: element automatic differentiation, constitutive automatic differentiation, and colored sparse finite-difference recovery.

We separate validation, derivative-route comparison, and parallel-solver behavior. The hyperelastic benchmark includes a serial companion comparison against JAX-FEM on a matched mesh and prescribed displacement schedule; all predefined agreement quantities are below the 5% threshold, and the displacement-field and energy differences are below 10^{-3} under the comparison contract. For the three-dimensional heterogeneous plasticity benchmark, direct-branch source-observable agreement of the endpoint surrogate is tight, with relative differences 3.877×10^{-5} in work, 3.517×10^{-3} in displacement, and 8.720×10^{-3} in deviatoric-strain norm. A separate fixed-load reference-operator diagnostic satisfies the prescribed $u_{\max}(\lambda_{\mathrm{sr}})$ and endpoint-displacement checks. These results support the implemented endpoint-surrogate workflow for the tested cases, but they do not claim path-consistent equivalence to an incremental elastoplastic history solve.

On a fixed $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, 8-rank plasticity derivative comparison, constitutive AD has the lowest median wall time among the three tested second-order routes while reaching the same terminal state as element AD and colored SFD. In the converged glued-bottom $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ scaling sweep, all single-node and multi-node CPU rows take the same nonlinear and Krylov work, while the single-node solver-total time drops from 17,890 s on one rank to 1110 s on 32 ranks. The topology optimization study complements this with a distributed design-and-mechanics workflow that remains operational at 768×384 resolution and 32 ranks. Within this scoped evidence, the contribution is a reusable nonlinear FEM toolset pattern: derivative-rich local models, sparse nonlinear solver policy, and distributed execution can be developed and assessed together across a heterogeneous family of energy-based problems.

1 Introduction

Nonlinear finite-element models in mechanics, scalar PDEs, and design optimization often combine three computational difficulties. The local model may be easiest to write as an energy or constitutive law whose derivatives are tedious to assemble by hand. The global problem then requires robust nonlinear globalization and sparse Newton–Krylov linear algebra. Finally, the implementation must remain comparable across reference formulations, derivative approximations, and distributed-memory solver choices. A useful computational toolset for this class of problems should therefore expose not only a modeling interface, but also the derivative route, assembly layout, nonlinear solver policy, and preconditioner design.

This paper studies that design point through an implemented toolset for nonlinear finite-element energy solves and optimization. The mainline path is a solver-centric JAX+PETSc stack: local energies and constitutive laws are written in JAX, while PETSc owns sparse vectors and matrices, nonlinear globalization, Krylov methods, and preconditioners. The benchmark suite also contains pure JAX and FEniCS references where those formulations are available, and it uses external or source-family comparisons only under narrow contracts. Thus the paper is not a claim that one framework dominates all finite-element workflows. It is a focused study of how local automatic differentiation, sparse assembly, nonlinear globalization, and distributed linear algebra can be kept in one documented nonlinear FEM workflow.

Recent open-source finite-element and PDE-optimization frameworks cover different parts of this workflow. High-level finite-element systems such as FEniCS and `DOLFINx` emphasize expressive variational modeling together with compiled kernels and parallel assembly [1, 2]. PDE-level adjoint frameworks such as `dolfin-adjoint`, `pyadjoint`, and `cashocs` automate adjoint-centered optimization workflows and PDE-level sensitivity automation [3–5]. JAX-native or JAX-connected workflows, including JAX-FEM, second-order implicit-differentiation benchmarks for finite-element differentiable physics, AutoPDEx, JAX-CPFEM, the Firedrake–JAX bridge, and recent `FEniCSx` external-operator work, push automatic differentiation deeper into constitutive modeling and differentiable simulation [6–11]. The JetSCI preprint is especially close in architecture because it also couples JAX-local finite-element kernels to PETSc for distributed sparse solves [12]. Recent topology-optimization implementations such as `FEniTop` show that compact parallel design-mechanics workflows can also be built on modern `FEniCSx` infrastructure [13].

1.1 Contributions

The contribution is a methods-and-software study of a nonlinear FEM toolset, organized around three solver components:

1. **Nonlinear solver policy.** We compare line-search, trust-region, and hybrid globalization choices on representative nonlinear finite-element benchmarks, and we keep the reported conclusions tied to the tested problem contracts.
2. **Derivative and assembly routes.** We implement and compare element automatic differentiation, constitutive automatic differentiation, and colored sparse finite-difference recovery inside the same sparse assembly framework.
3. **Sparse linear solver realization.** We connect the nonlinear and derivative choices to PETSc Krylov methods and preconditioners, including Hypre, GAMG, and problem-specific polynomial multigrid policies used in the larger mechanics and topology runs.

Together, these components make derivative construction, sparse assembly, globalization, and preconditioning part of one numerical-method choice rather than separate implementation details.

We evaluate these components on six benchmark families: p-Laplace, scalar Ginzburg–Landau, hyperelasticity, Plasticity2D, Plasticity3D, and topology optimization. The scalar problems isolate formulation and derivative agreement. Hyperelasticity adds finite-strain mechanics and a narrow JAX-FEM companion comparison. Plasticity2D and Plasticity3D test nonsmooth constitutive structure, source-observable agreement, and multigrid policy. Topology optimization tests a distributed design-and-mechanics loop. The geomechanics context follows the finite-element strength-reduction and 3D slope-stability literature [14–18], but the plasticity object studied here is explicitly an algorithmic endpoint surrogate, not a claim of path-consistent incremental-history equivalence.

1.2 State of the Art and Positioning

For the purposes of this paper, the most relevant comparison axes are the modeling layer, the scope of differentiation, the form in which second-order information enters the workflow, the parallel execution model, and the benchmark families closest to our suite. Table 1 summarizes that comparison using conservative, source-backed descriptors.

Three comparisons matter most for the narrative that follows. First, relative to high-level FEM and adjoint ecosystems [1–5], the present toolset moves the center of gravity from PDE-level adjoint-centered automation toward distributed nonlinear solve policy and multiple second-order routes inside the same sparse solver stack. Second, relative to JAX-native mechanics and differentiable-simulation frameworks [6–9], the main distinction is that the implementation here centers PETSc-owned sparse MPI data structures and preconditioned Newton solves, whereas the cited JAX-native frameworks emphasize JAX-level modeling, local differentiation, GPU execution, second-order implicit differentiation, or differentiable-physics workflows. Third, relative to recent bridge and hybrid papers that combine JAX-style differentiation with established FEM or solver environments [10–12], the paper's emphasis is on extending that bridge-style differentiation story into benchmarked PETSc sparse assembly, Krylov solves, and nonlinear globalization for benchmark families that include topology optimization and 3D Mohr–Coulomb-type mechanics.

The resulting positioning statement is intentionally modest. The toolset is not presented as a general differentiable-PDE platform, nor as a replacement for the high-level FEM and adjoint ecosystems cited above. It is presented as a JAX+PETSc workflow for distributed nonlinear energy minimization and equilibrium solves in which derivative fidelity, sparse assembly, nonlinear globalization, and preconditioner policy are all part of the experimental design rather than merely implementation detail.

The paper is organized as follows. Section 2 deepens the literature context behind the summary above. Section 3 describes the common finite-element energy notation, derivative routes, nonlinear globalization, and sparse derivative recovery. Section 4 explains how these ideas are realized through pure JAX, FEniCS, and JAX+PETSc paths. Section 5 defines the benchmark suite, with each benchmark separated into literature model and implemented discrete problem. Section 6 then reports validation and external-comparison evidence, deliberately separating hyperelastic JAX-FEM agreement from the Plasticity3D source-family diagnostic ladder. Section 7 reports timing, scaling, derivative-route, and discretization-comparison evidence, followed by supporting solver evidence and a discussion of limitations.

Family	Modeling	Differentiation	Second-order route	Parallel	Closest overlap
FEniCS DOLFINx [1, 2]	High-level variational forms	Problem-specific; AD is not the central claim	Manual or application-specific	Distributed FEM assembly and solve	Elliptic and finite-strain mechanics
dolfin-adjoint pyadjoint cashocs [3–5]	High-level PDE plus optimization loop	Adjoint-based first-order sensitivities	Not the advertised main path	MPI via the host FEM stack	PDE control, shape, and topology
JAX-FEM Xue 2026 AutoPDEx [6–8]	JAX-native PDE / FE programs	Program-level forward and reverse AD	Higher-order JAX and implicit Hessian-vector derivatives	JAX execution; PETSc optional in AutoPDEx	Nonlinear mechanics, inverse design, and FE differentiable physics
JetSCI [12]	JAX local discretizations plus PETSc sparse solves	JAX-differentiated discretization kernels	Differentiable simulation kernels with PETSc solves	JAX/GPU within node; PETSc MPI across nodes	Heterogeneous micromechanics
Firedrake–JAX FEniCSx ext. ops [10, 11]	Host FEM stack plus AD bridge	Tangent, adjoint, or local constitutive AD	Local-operator derivatives, not full sparse Hessians	Host framework parallel back end	Parameterized PDEs and constitutive models
JAX-CPFEM [9]	JAX-native crystal-plasticity FEM	Differentiable constitutive simulator	AD constitutive derivatives	GPU-oriented execution	Crystal plasticity
FEniTop [13]	FEniCSx topology code	Sensitivity-based design updates	Not a second-order solver paper	Parallel FEniCSx workflow	2D and 3D topology optimization
This work	JAX local energies plus PETSc sparse solvers	Element AD, constitutive AD, and colored SFD	Local Hessians/tangents or sparse recovery	PETSc MPI vectors, matrices, SNES, and multigrid	p-Laplace, GL, hyperelasticity, plasticity, topology

Table 1: State-of-the-art positioning of the framework families most relevant to this paper. The columns record source-supported workflow properties rather than subjective scores.

2 Related Work

The literature most relevant to this manuscript can be organized around four questions. How are finite-element models and PDE-constrained optimization workflows expressed? Where does automatic differentiation enter the finite- element computation? Which mechanics and topology benchmarks define the scientific problem classes? Which sparse nonlinear solver infrastructure makes the resulting methods scalable? We use these questions to position the present toolset without ranking the surrounding software ecosystems.

Finite-element and PDE-optimization automation. The original FEniCS project established the high-level finite-element programming model in which weak forms are expressed symbolically and compiled into efficient kernels [1]. The more recent `DOLFINx` preprint keeps that mathematical abstraction while moving to a data-oriented design with explicit support for modern parallelism, custom kernels, and direct access to lower-level data structures [2]. In PDE-constrained optimization, `dolfin-adjoint` showed that adjoints of high-level transient finite-element programs can be derived automatically from the symbolic problem description [3], and `pyadjoint` documents the FEniCS/Firedrake adjoint framework used to automate such workflows in practice [4]. The `cashocs` software paper extends this PDE-optimization thread to shape optimization, optimal control, topology optimization via level sets, and MPI-enabled workflows [5]. These works are the natural reference point for expressive finite-element and adjoint-centered automation. The present paper uses that context while focusing instead on second-order routes, nonlinear globalization, and sparse distributed solver policy.

Differentiable FEM and JAX–solver bridges. The broad AD background is surveyed by Baydin et al. [19], while JAX provides a program-transformation environment for higher-order derivatives, vectorization, and just-in-time compilation in scientific computing [20]. Within finite elements, JAX-FEM demonstrates inverse design, nonlinear constitutive modeling, and GPU-oriented differentiable workflows in a JAX-native 3D FE solver [6]. Xue's later finite-element differentiable-physics paper is a direct second-order comparator because it derives implicit Hessian-vector products and evaluates Newton–CG use of exact Hessian information on finite-element inverse-problem benchmarks [7]. AutoPDEx broadens the JAX-native PDE picture with nonlinear minimizers, implicit differentiation, and optional PETSc integration [8]; JAX-CPFEM adds differentiable crystal plasticity and GPU execution in a recent mechanics workflow [9].

Bridge architectures ask a complementary question: how can JAX-style differentiation be combined with established finite-element or sparse-solver environments? The Firedrake–JAX bridge uses tangent-linear and adjoint equations derived from the high-level PDE representation, avoiding differentiation through low-level solver iterations [10]. The `FEniCSx` external-operator work of Latyshev et al. [11] uses algorithmic automatic differentiation for general constitutive models inside FEniCSx, including a Mohr–Coulomb example. The JetSCI preprint is especially close architecturally because it uses JAX for local finite-element discretization and PETSc for robust distributed sparse solves in heterogeneous micromechanics [12]. These works motivate the JAX-local/PETSc-global design point. The distinction here is that we compare element AD, constitutive AD, and colored SFD inside one nonlinear energy-minimization suite spanning the benchmark families below.

Mechanics and topology benchmark context. The plasticity benchmarks use Mohr–Coulomb-type slope-stability problems, so the relevant literature separates continuum plasticity, strength-reduction practice, implementation lineage, and the endpoint surrogates studied here. Davis' plasticity treatment and the finite-element strength-reduction studies of Tschuchnigg et al. [14] provide the Davis-style reduction context used in the paper. For constitutive integration, the incremental reference frame is return mapping with consistent tangents [21, 22], with Mohr–Coulomb-specific nonsmooth return operators treated by Sysala et al. [15]. The recent Sysala slope-stability line documents variational and optimization viewpoints for shear strength reduction [16, 17] and a 3D continuation/iterative-solver source family [18]. The MATLAB/Octave implementation work of Čermák et al. [23] documents an efficient 2D/3D elastoplastic finite-element implementation lineage for this problem class. We use this literature to position the benchmark family; the numerical claims remain limited to the implemented endpoint surrogate and the reported comparisons.

For topology optimization, the classical SIMP and educational references supply the compliance, filtering, and continuation ideas used throughout the field [24–27]. A relevant modern software reference is FEniTop, which documents a 2D/3D FEniCSx topology-optimization implementation with parallel support [13]. It does not duplicate the reduced design objective used here, but it contextualizes compact parallel topology optimization in the modern FEniCSx ecosystem.

Sparse derivative recovery and scalable nonlinear infrastructure. When exact second-order information is too expensive, sparse derivative recovery via graph coloring remains the classical alternative. Curtis, Powell, and Reid gave the foundational sparse-Jacobian viewpoint [28], while Coleman and Moré extended the approach to sparse Jacobian and Hessian estimation with graph-coloring structure [29, 30]. PETSc supplies the distributed vectors, matrices, Krylov methods, nonlinear solvers, and multigrid infrastructure needed to turn those derivative objects into scalable workflows [31, 32]. In this paper, PETSc is not treated as a mere back-end library: ownership layout, Krylov policy, nonlinear globalization, and preconditioner design are all part of the scientific comparison.

Against that background, the niche of the present toolset can be stated precisely. It sits between high-level differentiable modeling and scalable sparse nonlinear solver engineering. The contribution is not that it replaces the ecosystems above, but that it documents one specific point in the contemporary design space: implemented distributed nonlinear FEM workflows that combine JAX-based local differentiation, multiple second-order routes, and PETSc-based sparse execution on benchmark families ranging from scalar nonlinear diffusion to topology optimization and 3D Mohr–Coulomb-type mechanics.

3 Methodology

We use a common finite-element notation across all benchmark families. The continuous domain is $\Omega \subset \mathbb{R}^d$, the conforming trial space is $V_h \subset V$, the state vector is $u_h \in V_h$, and the element-restricted state on element e is u_e. When a family is posed as an energy or reduced objective minimization, we write its discrete functional as $\mathcal{J}_h(u_h)$; when a family is posed as a sequence of equilibrium solves or by a scalar equilibrium surrogate, we write the associated scalar functional as $\Pi_h(u_h)$. This distinction records the modeling role of the functional, not a difference in the sparse assembly pattern. When an algorithm is stated independently of a benchmark family, we denote its merit functional by $\mathcal{F}$. Hyperelasticity is the only family in which $J = \det F$ denotes the deformation Jacobian, so we avoid the bare symbol J for generic objectives elsewhere in the paper. For finite-element labels, $P_k(L_\ell)$ denotes degree-k Lagrange elements on mesh level L_ℓ; problem-specific sections define the mesh hierarchy behind those levels when it matters for interpretation.

At the discrete algebraic level, the common sparse-assembly task is therefore one of two closely related forms:

$$\mathcal{J}_h(u_h) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(u_e) - f_{\mathrm{ext}}^T u_h, \tag{1}$$

or

$$\Pi_h(u_h) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(u_e) - f_{\mathrm{ext}}^T u_h, \tag{2}$$

where Φ_e may represent a scalar diffusion energy, a phase-field potential, a finite-strain stored-energy density, an algorithmic scalarized plasticity surrogate, or the reduced design objective used in topology optimization. The benchmark sections specialize equations (1) and (2) to each family and define any additional symbols locally.

3.1 Finite-element and derivative structure

The main design choice in the toolset is that problem definitions should remain energy-centered whenever possible. This gives three derivative routes:

1. **element AD:** differentiate the full scalar element contribution $\Phi_e(u_e)$ with respect to the element DOF vector;
2. **constitutive AD:** differentiate a quadrature-point energy density $\psi(\varepsilon_q)$, then assemble $r_e = \sum_q w_q B_q^T \sigma_q$ and $K_e = \sum_q w_q B_q^T C_q B_q$;
3. **colored SFD:** recover sparse Hessian information through directionally probed gradients/HVPs and graph coloring.

Element AD is the most literal route and is often the cleanest mathematically. Constitutive AD preserves the same scalar-energy source of truth but changes the differentiation boundary from the full element vector to the local strain state. This is especially effective for high-order plasticity, where the full element Hessian is often much more expensive than the 6×6 constitutive tangent. Colored SFD is used when exact second-order information is too costly or unnecessary.

3.2 Newton–Krylov globalization

The nonlinear solves are based on Newton or quasi-Newton updates, but they are not all globalized in the same way. Scalar benchmarks primarily use line-search Newton. Hyperelasticity uses a trust-region method with optional post-step line search. Some benchmark configurations use a hybrid trust-region/line-search algorithm in which a reduced trust-region model provides the direction, while a bounded line search chooses the accepted step. The line-search and trust-region ingredients follow standard globalization ideas [33, Chs. 2–4]; the particular hybrid coupling used in some benchmark configurations is a problem-specific combination of those textbook ingredients.

Algorithm 1 Hybrid trust-region / line-search Newton

Given x_0, trust radius Δ_0, tolerances, and callbacks for energy, gradient, and Hessian solves/matvecs
for $k = 0, 1, \ldots$ **do**
    Assemble $g_k = \nabla \mathcal{F}(x_k)$
    if convergence criteria are satisfied **then**
        terminate
    end if
    Compute a Newton-like direction from $\nabla^2 \mathcal{F}(x_k) p = -g_k$
    if trust region is enabled **then**
        Build a small reduced model in the span of Newton and gradient directions
        Solve the reduced trust subproblem with radius Δ_k
        Optionally compare against a pure gradient trust step
        Restrict the line-search interval so the accepted step stays in the trust ball
    else
        Use the Newton direction directly
    end if
    Perform one-dimensional search on the chosen direction
    Compute actual and predicted reduction
    if trust region is enabled **then**
        accept/reject using $\rho = \text{ared}/\text{pred}$ and update Δ_k
    else
        accept only if the merit function decreases
    end if
end for

For the principal 3D plasticity path reported in this paper, the chosen policy is simpler: Newton with Armijo backtracking and a PMG-preconditioned flexible-GMRES linear solve [33, Sec. 3.1]. This is the baseline workflow used for the converged 3D plasticity scaling study reported later in Section 7.7.

Algorithm 2 Armijo line search used in Newton solves

Given state x, direction p, initial step α_0
Evaluate $\mathcal{F}(x)$ and $g = \nabla \mathcal{F}(x)$
for $m = 0, 1, \ldots, m_{\max}$ **do**
    $\alpha \leftarrow \alpha_0 \beta^m$
    $x_{\text{trial}} \leftarrow x + \alpha p$
    if $\mathcal{F}(x_{\text{trial}}) \leq \mathcal{F}(x) + c_1 \alpha g^T p$ **then**
        accept α
        return
    end if
end for
fall back to the smallest admissible step or to gradient-safe logic

3.3 Colored sparse finite differences

Sparse derivative recovery is built around the observation that if columns that do not interfere in the sparsity pattern are perturbed together, one gradient or HVP evaluation can recover multiple columns of the Jacobian or Hessian [28, 30]. The local SFD path uses distance-2 coloring on the Hessian sparsity graph, which is especially effective when combined with overlap-domain assembly and owned-plus-ghost PETSc layouts.

Algorithm 3 Distributed colored Hessian recovery

Build or import local sparsity information for the free-DOF graph
Compute a distance-2 coloring of the Hessian pattern
for each color group c **do**
    Build the combined probe vector v_c
    Evaluate one gradient difference or HVP for v_c
    Scatter recovered entries into owned sparse storage
end for
Assemble the global PETSc matrix from owned COO/CSR entries

3.4 Constitutive autodiff for 3D plasticity

The main Plasticity3D result in this paper uses constitutive AD. Rather than applying second-order AD to the full P_4 element vector, we define one scalar Mohr–Coulomb branch potential $\Phi_q(\varepsilon_q, m_q)$ per quadrature point, where m_q collects the local material parameters. We differentiate this scalar potential with respect to the six engineering-strain components and assemble the element tangent from the resulting stress and constitutive tangent. This keeps the scalar-energy contract intact while reducing the cost of branchwise second-order information on high-order tetrahedra.

Algorithm 4 Constitutive AD assembly for high-order plasticity

for each local element e **do**
    for each quadrature point q **do**
        Evaluate strain $\varepsilon_q = B_q u_e$
        Evaluate scalar energy density $\Phi_q(\varepsilon_q)$
        Obtain stress $\sigma_q = \partial \Phi_q / \partial \varepsilon_q$
        Obtain constitutive tangent $C_q = \partial^2 \Phi_q / \partial \varepsilon_q^2$
    end for
    Assemble $r_e = \sum_q w_q B_q^T \sigma_q$
    Assemble $K_e = \sum_q w_q B_q^T C_q B_q$
end for

The key methodological point is that automatic differentiation is applied directly to the implemented scalar branch potential. The implementation does not switch to a hand-derived constitutive tangent. Away from branch switching sets this yields branchwise exact derivatives of the implemented surrogate; at switching sets the model remains nonsmooth, so the resulting tangents should be read as branchwise surrogate derivatives rather than globally smooth continuum derivatives.

4 Implementation

The implementation is organized around three solver paths with different scientific roles.

Pure JAX reference paths. These are compact differentiable realizations of selected benchmark energies. They are useful for formulation checks, serial references, and AD diagnostics. They are not intended to be the primary scalable route for the largest distributed benchmarks. Their main scientific role in this paper is to show that the same energy formulations can also be expressed without sparse distributed infrastructure.

FEniCS reference paths. The FEniCS implementations remain valuable on families where they exist, especially p-Laplace, Ginzburg–Landau, and hyperelasticity. They provide a trusted high-level FEM reference and help distinguish modeling issues from solver-stack issues. In this paper they are used mainly as comparison baselines rather than as the mainline distributed path.

Mainline JAX+PETSc path. This is the central solver realization of the paper. It combines JAX-based local physics with PETSc distributed vectors, matrices, Krylov solvers, and preconditioners. Problem-specific kernels stay in JAX; sparse distributed assembly, nonlinear globalization, and linear algebra stay in PETSc.

4.1 Autodiff modes

The JAX+PETSc stack supports several second-order modes. The precise available choices depend on the benchmark family, but the common logic is:

- **element AD**: exact local Hessians from differentiating the full element energy;
- **constitutive AD**: exact local constitutive tangents assembled from quadrature-point energy densities;
- **local SFD / colored SFD**: sparse approximate Hessian recovery when exact second-order information is not the appropriate cost tradeoff.

This split is scientifically important. It lets us compare exact second-order information against carefully structured sparse approximations without changing the rest of the nonlinear solver stack. In other words, the paper is not only a comparison of frameworks, but also a comparison of derivative strategies inside one common solver infrastructure.

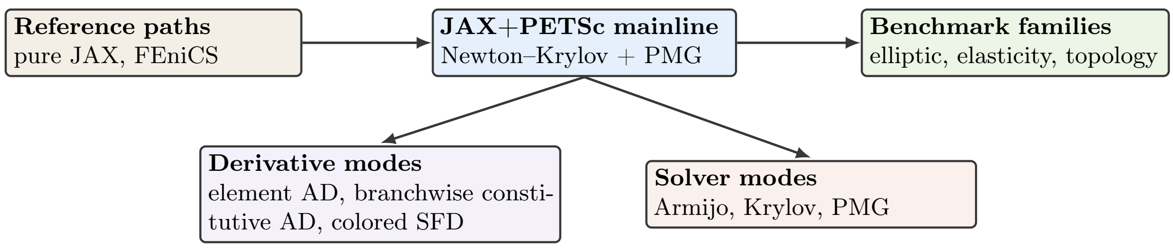


Figure 1: Software structure of the manuscript. The JAX+PETSc stack is the mainline distributed path; pure JAX and FEniCS remain reference paths where matching benchmark formulations exist.

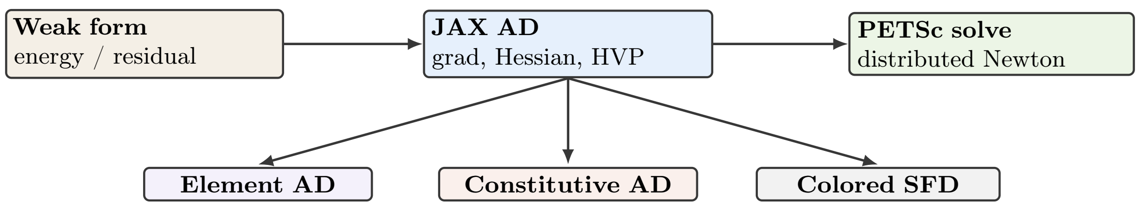


Figure 2: Derivative pathways from scalar energies and quadrature-point potentials to distributed PETSc nonlinear solves.

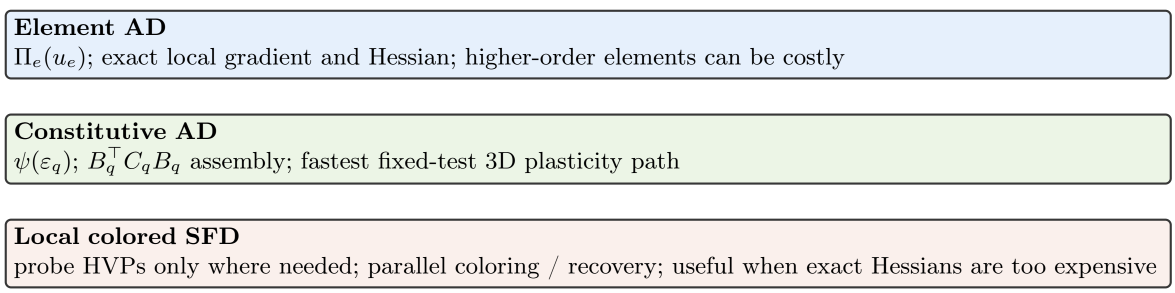


Figure 3: Second-order derivative modes exposed in the JAX+PETSc stack. The important engineering choice is that the source of truth remains an energy or energy-density expression even when the tangent path changes from element AD to constitutive AD or colored SFD.

4.2 Linear solvers and preconditioners

The benchmark families use different Krylov and preconditioner combinations because the linear systems are qualitatively different:

- **p-Laplace:** mostly CG with Hypre, with exact element Hessians or local-SFD Hessians in the JAX+PETSc path.
- **Ginzburg–Landau:** GMRES with Hypre, reflecting the more indefinite linearization.
- **Hyperelasticity:** trust-region outer iterations with GAMG-preconditioned Krylov solves in the main benchmark and globalization rows, and same-hierarchy PMG with redundant LU/MUMPS coarse solves in the larger fixed-work scaling profile.
- **Plasticity (2D):** deep-tail multigrid hierarchies whose bottlenecks are dominated by the top smoother and repeated Krylov work.
- **Plasticity3D:** same-mesh polynomial multigrid hierarchies, $P_4 \to P_2 \to P_1$ or $P_2 \to P_1$ depending on degree. The auxiliary $\lambda_{\mathrm{sr}} = 1.0$ scaling and degree/resolution endpoint probes use FGMRES with same-mesh PMG and a CG–Hypre coarse solve, whereas the P_2 globalization and derivative-route comparisons and the converged $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ scaling use a redundant LU/MUMPS coarse profile.
- **Topology:** flexible GMRES with GAMG for the mechanics step, coupled to a separate distributed design update.

The comparison material later in the manuscript also includes a source-family PMG-shell variant and alternative Krylov/preconditioner diagnostics. Those are not the primary defaults, but they are informative because they reveal how much of the final performance difference comes from derivative paths, preconditioners, or outer solver logic.

4.3 Distributed assembly and coloring

The JAX+PETSc implementation uses PETSc-owned ranges, local-plus-ghost layouts, and owned-row sparse assembly. This matters for both exact and approximate second-order information. Exact element or constitutive assembly can write directly into local sparse buffers. Colored SFD requires an additional graph-coloring step, but once the coloring is known the probe evaluations can still be carried out in a rank-local manner with only scalar reductions or owned sparse insertion in the hot loop.

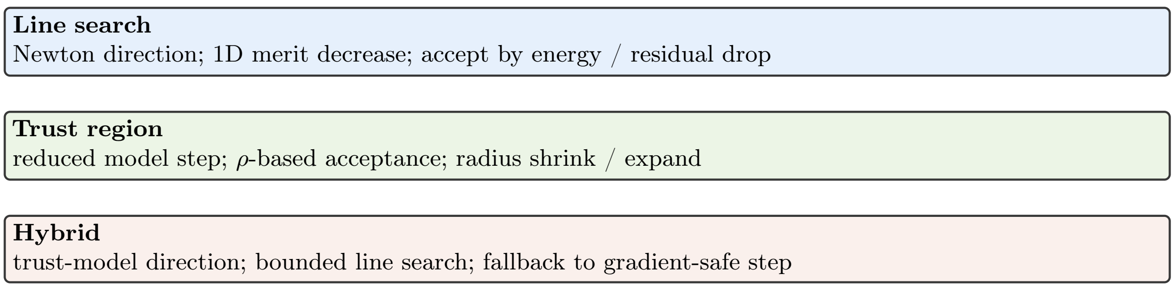


Figure 4: Globalization choices used across the benchmark suite, from pure line search to trust-region and hybrid strategies.

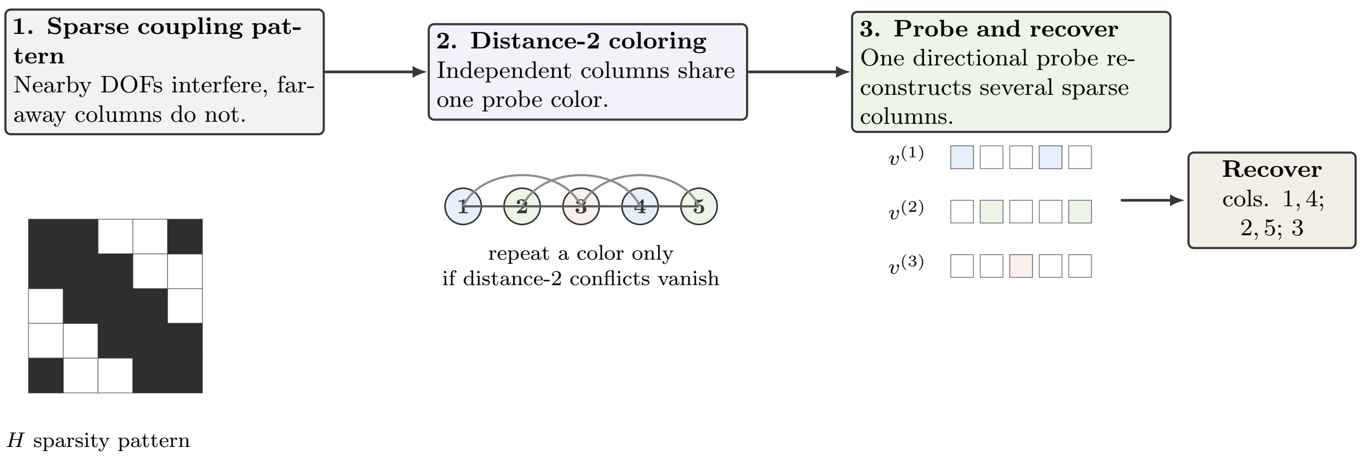


Figure 5: Schematic view of the colored sparse-recovery logic used by the local SFD Hessian path: a sparsity pattern, a distance-2 coloring, and the corresponding grouped probes used for recovery.

Family	FEniCS	pure JAX	JAX+PETSc	Advanced solver / derivative features
p-Laplace	yes	yes	yes	element AD and local colored SFD
Ginzburg–Landau	yes	no	yes	element AD and local colored SFD
Hyperelasticity	yes	yes	yes	trust-region element AD
Plasticity2D	no	no	yes	scalarized endpoint potential and same-mesh PMG
Plasticity3D	no	no	yes	constitutive AD, element AD, and same-mesh PMG
Topology optimization	no	yes	yes	distributed design updates and PETSc mechanics

Table 2: Implementation capability matrix across the benchmark families. "Advanced solver / derivative features" summarizes the derivative routes, nonlinear policies, or coupled solver features that are scientifically interesting in each family.

Table 2 summarizes the implemented surface. The most important structural point is that the JAX+PETSc distributed story is always the JAX+PETSc column, even when a family also has FEniCS or pure JAX references.

5 Benchmark Problems

The benchmark suite is intentionally heterogeneous. The central claim of the paper is not that one solver policy dominates every nonlinear PDE family, but that one differentiable sparse-solver stack can cover scalar elliptic problems, phase-field-type energies, finite-strain mechanics, elastoplastic slope stability, and reduced topology optimization without switching conceptual frameworks.

Family	Grid / mesh	Solve contract	Compared paths	Main difficulty
p-Laplace	L_9	Newton + line search	FEniCS, pure JAX, JAX+PETSc	nonlinear elliptic solve with exact sparse Hessians
Ginzburg–Landau	L_9	Newton + line search	FEniCS, JAX+PETSc	indefinite local curvature from the double well
Hyperelasticity	L_4, 24 steps	trust-region path	FEniCS, pure JAX, JAX+PETSc	nonconvex large-deformation mechanics
Plasticity2D	$P_4(L_5)$–$P_4(L_7)$	endpoint or fixed work	JAX+PETSc only	same-mesh PMG and nonlinear tail behavior
Plasticity3D	$P_1/P_2/P_4$	configuration-specific	constitutive and reference PMG variants	heterogeneous 3D Mohr–Coulomb with constitutive AD
Topology	768×384	stall-stop continuation	pure JAX, JAX+PETSc	distributed design-mechanics coupling

Table 3: Benchmark families, stopping rules, and principal comparison paths. The later numerical section revisits the same families from the timing and scaling viewpoint.

Family	FEniCS	pure JAX	Notes
p-Laplace	yes	yes	All three stacks exist on representative reported cases.
Ginzburg–Landau	yes	no	FEniCS and JAX+PETSc form the reported comparison.
Hyperelasticity	yes	yes	pure JAX is a serial formulation reference only.
Plasticity2D	no	no	The reported solver path is JAX+PETSc only.
Plasticity3D	no	no	Reference-formula assembly exists as supporting comparison, not as a reference path.
Topology	no	yes	Parallel fine-grid path is JAX+PETSc; pure JAX remains the serial design reference.

Table 4: Availability of FEniCS and pure JAX references. Reference paths are used where documented implementations exist for the same benchmark family; absent cells are left absent rather than filled with speculative baselines.

5.1 p-Laplace

The literature model is the standard stationary p-Laplace Dirichlet problem and its associated energy formulation [34, Ch. 2]. The unknown is a scalar field $u : \Omega \to \mathbb{R}$ and the continuous energy is

$$\mathcal{E}(u) = \int_\Omega \frac{1}{p}|\nabla u|^p \, dx - \int_\Omega f u \, dx, \tag{3}$$

whose Euler–Lagrange equation is

$$-\nabla \cdot \left(|\nabla u|^{p-2}\nabla u\right) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{4}$$

The weak problem is: find $u \in W_0^{1,p}(\Omega)$ such that

$$\int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall v \in W_0^{1,p}(\Omega). \tag{5}$$

The implemented benchmark used in this paper takes $\Omega = (0,1)^2$, exponent $p = 3$, forcing $f = -10$, and a conforming P_1 triangular hierarchy. The discrete problem is

$$u_h \in V_h \subset W_0^{1,p}(\Omega): \qquad u_h = \arg\min_{v_h \in V_h} \mathcal{E}_h(v_h), \tag{6}$$

where V_h is the chosen conforming finite-element subspace. This is the cleanest benchmark in the paper for checking exact gradients, exact Hessians, and sparse-recovery alternatives against each other without constitutive or geometric complications.

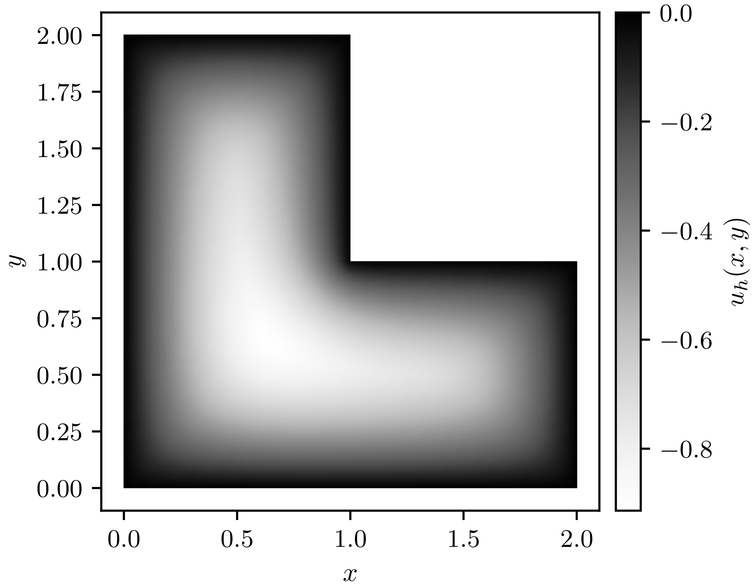


Figure 6: Representative p-Laplace state on the selected hierarchy.

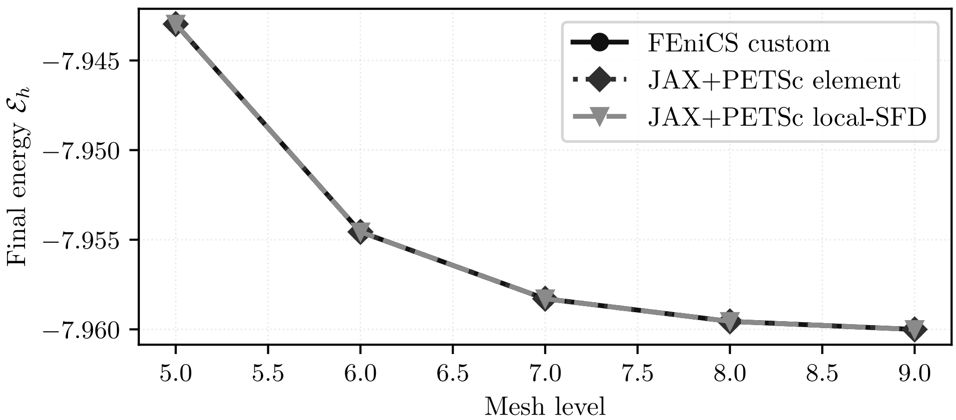


Figure 7: Final discrete energies from equation (6) across the selected mesh hierarchy. The element-AD and local-SFD JAX+PETSc paths are essentially energy-consistent with the FEniCS reference.

Path	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS custom Newton	−7.942,968,719	5	15	0.0782
JAX+PETSc element AD	−7.942,968,719	6	17	0.0798
JAX+PETSc colored SFD	−7.942,968,711	6	6	0.194

Table 5: Representative final discrete energies from equation (6) on the showcase case.

The within-benchmark picture is therefore one of formulation agreement: figure 7 and table 5 show that the implemented routes converge to the same discrete energy from equation (6). Scaling and timing for this family are reported in Section 7.3.

5.2 Ginzburg–Landau

We cite Ginzburg and Landau [35] for the historical origin of Ginzburg–Landau free-energy modeling only. The benchmark used here is a scalar real-valued double-well problem on $\Omega = [-1,1]^2$ with homogeneous Dirichlet boundary conditions and energy

$$\mathcal{E}(u) = \int_\Omega \frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{4}(u^2-1)^2\,dx, \qquad \varepsilon = 10^{-2}. \tag{7}$$

Its strong form is

$$-\varepsilon\Delta u + u(u^2-1) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{8}$$

and the weak problem is: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \varepsilon\nabla u\cdot\nabla v + (u^2-1)uv\,dx = 0 \qquad \forall v \in H_0^1(\Omega). \tag{9}$$

The discrete problem is a conforming P_1 stationary problem on the same triangular hierarchy:

$$u_h \in V_h: \qquad D\mathcal{E}_h(u_h)[v_h] = 0 \qquad \forall v_h \in V_h. \tag{10}$$

Relative to p-Laplace, the additional difficulty here is not geometry but indefinite local curvature from the double-well potential in equation (7). All compared routes use the same initial state

$$u_0(x,y) = \sin\left(\frac{\pi(x-1)}{2}\right)\sin\left(\frac{\pi(y-1)}{2}\right),$$

restricted to the free degrees of freedom, so the nonconvex branch selection is part of the benchmark contract.

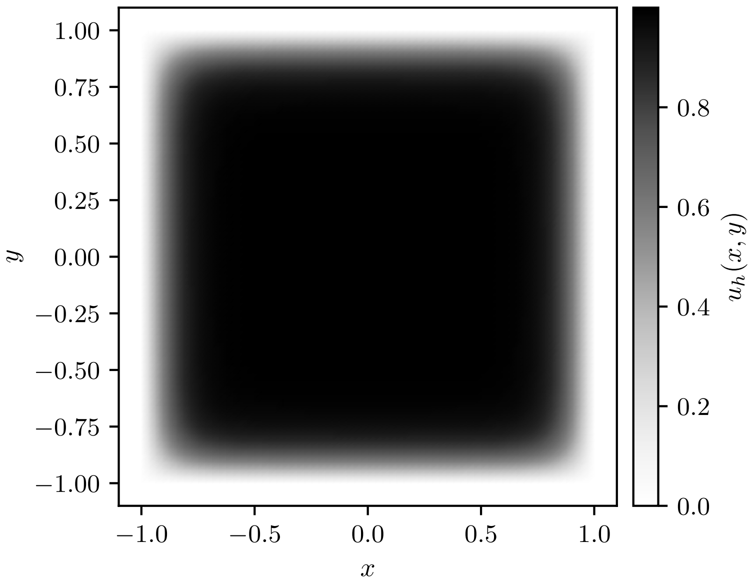


Figure 8: Representative Ginzburg–Landau state on the selected hierarchy.

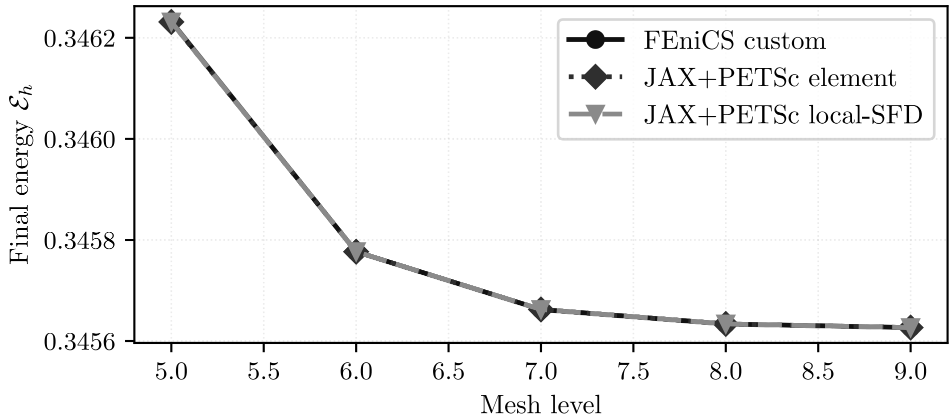


Figure 9: Final discrete energies associated with the stationary problem equation (10) across mesh levels. The element-AD and local-SFD JAX+PETSc routes remain nearly coincident with the FEniCS reference.

Path	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS custom Newton	0.346,232,365,1	8	33	0.113
JAX+PETSc element AD	0.346,231,443,8	8	33	0.0949
JAX+PETSc colored SFD	0.346,231,443,8	8	33	0.247

Table 6: Representative final discrete energies associated with equation (10) on the showcase case.

The local picture is again one of formulation consistency, now for a more solver-sensitive nonconvex energy. Strong scaling and solver-time comparisons for this family are reported in Section 7.4.

5.3 Hyperelasticity

The literature model is a standard finite-strain hyperelastic equilibrium problem written in terms of deformation map $y(X) = X + u(X)$, deformation gradient $F = \nabla y$, Jacobian $J = \det F$, and first Piola stress $P = \partial W/\partial F$ [36, Chs. 4–6]. The implemented benchmark used in this paper is a cantilever problem with the compressible Neo-Hookean-type density

$$\Pi(y;\eta) = \int_\Omega C_1 \big(\mathrm{tr}(F^T F) - 3 - 2\ln J\big) + D_1 (J-1)^2 \, dx, \tag{11}$$

on $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$ with $C_1 \approx 3.846 \times 10^7$ and $D_1 \approx 8.333 \times 10^7$. The left face is clamped, while the right face carries the prescribed rotating Dirichlet load path used throughout this benchmark. For a reference point $X = (X_1, X_2, X_3)$ on the right face and $N = 24$ reported load steps, the prescribed position at step k is

$$\bar{y}_k(X) = \big(X_1, \cos\eta_k\, X_2 + \sin\eta_k\, X_3, -\sin\eta_k\, X_2 + \cos\eta_k\, X_3\big), \qquad \eta_k = \frac{8\pi k}{N}.$$

The strong form is

$$-\nabla \cdot P(F) = 0 \quad \text{in } \Omega, \qquad y = \bar{y}(\eta) \quad \text{on } \Gamma_D, \qquad P(F)n = 0 \quad \text{on } \Gamma_N, \tag{12}$$

where $P = \partial W/\partial F$ is the first Piola stress. The weak form is: find $u \in V_{\bar{y}(\eta)}$ such that

$$\int_\Omega P(F(u)) : \nabla v \, dx = 0 \qquad \forall v \in V_0. \tag{13}$$

The discrete problem solved in the paper is the sequence of stationary points

$$u_h^{(k)} \in V_h : \qquad D\Pi_h(u_h^{(k)}; \eta_k)[v_h] = 0 \qquad \forall v_h \in V_{h,0}, \qquad k = 1, \ldots, 24, \tag{14}$$

with first-order tetrahedral vector elements and the benchmark-specific boundary data described above. This is the first family in which globalization is more scientifically important than derivative parity alone.

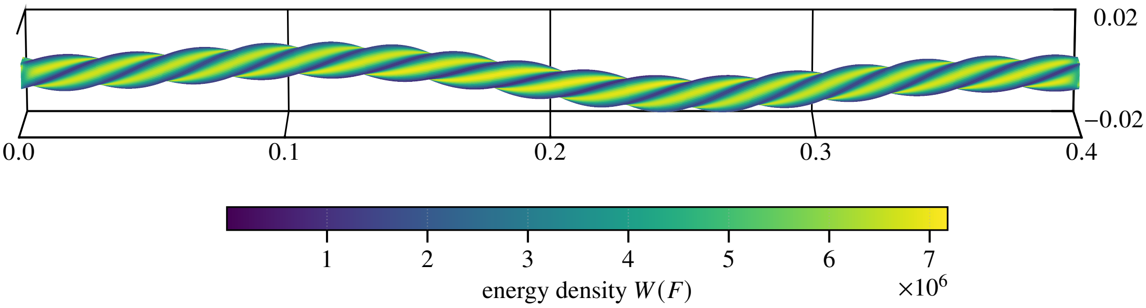


Figure 10: Representative finite-strain hyperelastic state, colored by the energy density in equation (11).

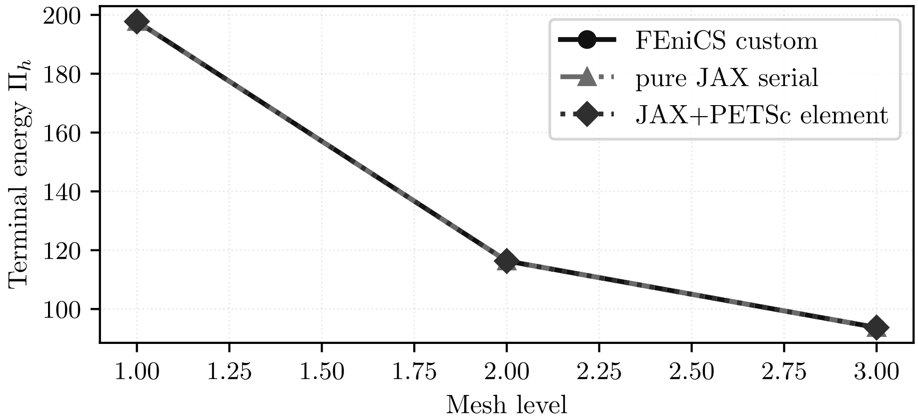


Figure 11: Final discrete energies along the stationary sequence equation (14) across the selected mesh hierarchy. The comparison is less about exact equality than about tracking the same 24-step trajectory consistently.

Path	Energy	Steps	Krylov iters	Wall time [s]
FEniCS custom Newton	197.775,074	24	8722	6.51
JAX+PETSc element AD	197.754,582	24	10595	9.36
serial JAX	197.749,509	24	2284	41.5

Table 7: Representative 24-step terminal energies along equation (14).

The within-benchmark takeaway is therefore a globalization result: figures 10 and 11 show the terminal deformed state and terminal energies, while the scaling evidence is deferred to Section 7.5. The separate validation section adds one serial companion comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with energy post-comparison under equation (11); it is included as an agreement-focused hyperelastic implementation comparison rather than as a broad framework ranking exercise.

5.4 Plasticity2D

The literature model is a quasi-static plane-strain Mohr–Coulomb slope stability problem posed as equilibrium together with a constitutive update and history variables [14–16, 21, 22, 37]. The unknown is the displacement field $u = (u_x, u_y) : \Omega \to \mathbb{R}^2$ on a homogeneous slope geometry. At increment $n+1$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = b^{n+1} \quad \text{in } \Omega, \qquad u^{n+1} = \bar{u}^{n+1} \quad \text{on } \Gamma_D, \qquad \sigma^{n+1} n = t^{n+1} \quad \text{on } \Gamma_N, \tag{15}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\, dx = \int_\Omega b^{n+1} \cdot v\, dx + \int_{\Gamma_N} t^{n+1} \cdot v\, ds \qquad \forall v \in V_0. \tag{16}$$

The runs reported here are endpoint solves, not path-consistent incremental histories. Accordingly, the benchmark uses the following scalarized discrete surrogate functional for differentiation and comparison:

$$\Pi_h^{\text{2D}}(u_h; \lambda_{\text{sr}}) = \sum_{e=1}^{n_{\text{el}}} \sum_{q=1}^{n_q} w_{eq}\, \Phi_{\text{MC,2D}}\big(B_{eq} u_e, \varepsilon_{eq}^{p,\text{old}}, E, \nu, c_{\lambda,eq}, \phi_{\lambda,eq}, \psi_{\lambda,eq}\big) - f_{\text{ext}}^T u_h, \tag{17}$$

where λ_{sr} denotes the strength-reduction factor, distinct from the Lamé parameter used later in three dimensions.

The benchmark uses a Davis-B-style reduction path to map the raw cohesion and angles to reduced values under strength reduction; with c_0, ϕ, ψ denoting the unreduced cohesion, friction angle, and dilation angle, respectively, the benchmark uses

$$
\begin{aligned}
c_1 &= c_0/\lambda_{\mathrm{sr}}, & \phi_1 &= \arctan(\tan\phi/\lambda_{\mathrm{sr}})\,, & \psi_1 &= \arctan(\tan\psi/\lambda_{\mathrm{sr}})\,, \\
\beta &= \frac{\cos\phi_1\cos\psi_1}{1-\sin\phi_1\sin\psi_1}, & c_\lambda &= \beta c_1, & \phi_\lambda &= \arctan(\beta\tan\phi_1)\,,
\end{aligned}
\tag{18}
$$

with ψ_λ defined analogously when the non-associated flow rule is activated [14, 16, 37]. The showcase endpoint runs in this paper use $\lambda_{\mathrm{sr}} = 1$ and $\psi = 0°$, so the reported states should be read as algorithmic endpoint surrogates rather than path-consistent history updates.

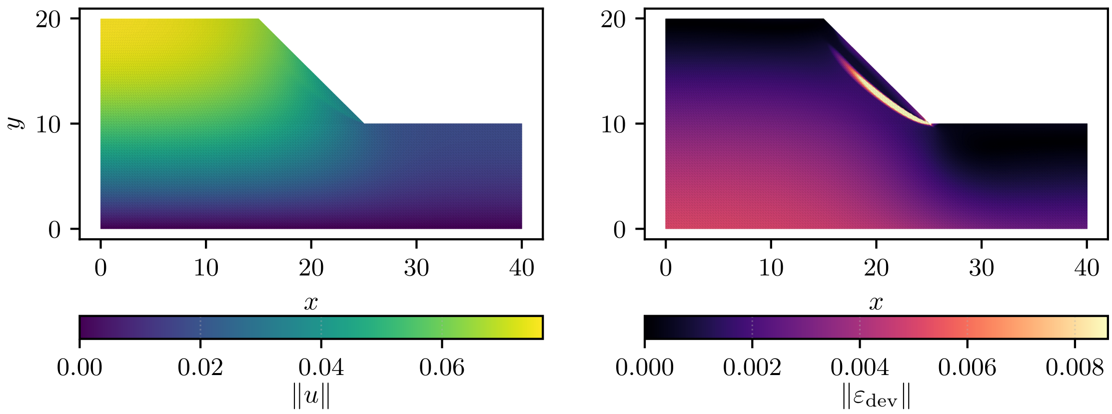


Figure 12: Representative Plasticity2D state: displacement magnitude and deviatoric strain derived from the scalar surrogate equation (17).

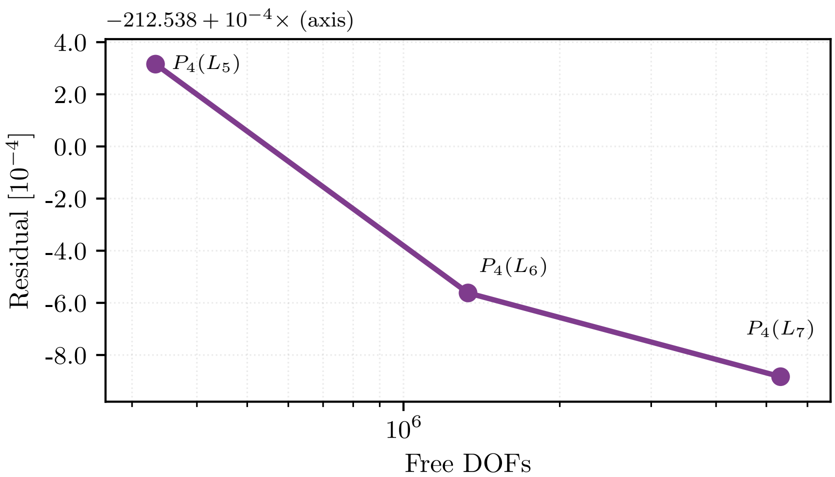


Figure 13: Resolution trend for the Plasticity2D family. The plotted terminal energies are evaluations of equation (17) on the curated $P_4(L_5)$ showcase and on the larger fixed-work $P_4(L_6)$ and $P_4(L_7)$ diagnostics, which are not used as endpoint-convergence evidence.

Case	Free DOFs	Energy	Wall time [s]	Note
$P_4(L_5)$	332,960	−212.538	147	curated showcase
$P_4(L_6)$	1,331,520	−212.539	147	fixed-work at 8 ranks
$P_4(L_7)$	5,325,440	−212.539	253	fixed-work at 16 ranks

Table 8: Representative Plasticity2D endpoint values from equation (17).

This family is where multigrid hierarchy design first becomes dominant in the paper. Only the curated $P_4(L_5)$ row is treated as a completed endpoint solve; the larger $P_4(L_6)$ and $P_4(L_7)$ rows are fixed-work diagnostics for solver behavior. Solver-engineering evidence for Plasticity2D is reported later in Sections 7.6 and 8.

5.5 Plasticity3D

The literature model is again quasi-static Mohr–Coulomb equilibrium with constitutive/history updates, now on a three-dimensional heterogeneous slope geometry [14–18, 21, 22, 37]. The unknown is a displacement field $u = (u_x, u_y, u_z) : \Omega \to \mathbb{R}^3$, gravity acts in the source y-direction. At increment $n+1$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = \rho g_y^{n+1} e_y \quad \text{in } \Omega, \qquad u_i^{n+1} = \bar{u}_i^{n+1} \quad \text{on } \Gamma_{D,i}, \qquad \sigma^{n+1} n = 0 \quad \text{on } \Gamma_N, \tag{19}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\, dx = \int_\Omega \rho g_y^{n+1} e_y \cdot v\, dx \qquad \forall v \in V_0. \tag{20}$$

The studied discrete model uses a heterogeneous source-family slope mesh hierarchy with same-mesh tetrahedral $P_1/P_2/P_4$ spaces, component-wise displacement boundary labels, and an additional glued-bottom condition that fixes every node with $|y| \le 10^{-12}$ in all three displacement components. We write $P_k(L_\ell)$ for polynomial degree k on mesh level ℓ and use $L_1, \ldots, L_4$ for the four displayed mesh levels. The figures below use the corrected glued-bottom $P_4(L_2)$ showcase at $\lambda_{\mathrm{sr}} = 1.55$ and the finished nine-case study at the same load factor. The $\lambda_{\mathrm{sr}} = 1.0$ strong-scaling study is kept later in the paper as separate auxiliary timing evidence. For those endpoint and continuation studies, we evaluate the following algorithmic scalar surrogate:

$$\Pi_h^{3\mathrm{D}}(u_h; \lambda_{\mathrm{sr}}) = \sum_{e=1}^{n_{\mathrm{el}}} \sum_{q=1}^{n_q} w_{eq}\, \Phi_{\mathrm{MC,3D}}(\varepsilon_{eq}(u_e), \bar{c}_{\lambda,eq}, \sin\phi_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{\mathrm{L},eq}) - f_{\mathrm{ext}}^T u_h, \tag{21}$$

where λ_{sr} is again the strength-reduction factor and λ_{L} is the Lamé parameter of the elastic predictor.

We use the engineering-strain ordering

$$\varepsilon = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \gamma_{xy} & \gamma_{yz} & \gamma_{xz} \end{bmatrix}^T, \tag{22}$$

which is first mapped to tensor shear before principal values are formed. The benchmark keeps the Davis-B reduction explicitly at quadrature points:

$$\begin{aligned} c_{1,q} &= c_{0,q}/\lambda_{\mathrm{sr}}, & \phi_{1,q} &= \arctan(\tan\phi_q/\lambda_{\mathrm{sr}}), & \psi_{1,q} &= \arctan(\tan\psi_q/\lambda_{\mathrm{sr}}), \\ \beta_q &= \frac{\cos\phi_{1,q}\cos\psi_{1,q}}{1 - \sin\phi_{1,q}\sin\psi_{1,q}}, & c_{\lambda,q} &= \beta_q c_{1,q}, & \phi_{\lambda,q} &= \arctan(\beta_q \tan\phi_{1,q}), \end{aligned} \tag{23}$$

with

$$\bar{c}_{\lambda,q} = 2c_{\lambda,q}\cos\phi_{\lambda,q}, \qquad \sin\phi_{\lambda,q} = \sin(\phi_{\lambda,q}), \tag{24}$$

which are the scalar constitutive parameters used by the 3D assembly path [14, 16, 37]. Here ϕ denotes friction angle and ψ denotes dilation angle; the paper uses those symbols only in that geomechanics sense.

The local constitutive surrogate follows the source-family piecewise branch structure. At one quadrature point we suppress the index q and write $s_\lambda := \sin\phi_\lambda$. Let $\varepsilon_1 \ge \varepsilon_2 \ge \varepsilon_3$ be the principal strains and let

$$I_1 = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}, \qquad q(\varepsilon) = \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right). \tag{25}$$

The elastic branch is

$$\Phi_{\mathrm{el}} = \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu q(\varepsilon), \tag{26}$$

and the branch-selection variables are

$$\begin{aligned} f_{\mathrm{tr}} &= 2\mu((1+s_\lambda)\varepsilon_1 - (1-s_\lambda)\varepsilon_3) + 2\lambda_{\mathrm{L}} s_\lambda I_1 - \bar{c}_\lambda, \\ \gamma_{\mathrm{sl}} &= \frac{\varepsilon_1 - \varepsilon_2}{1+s_\lambda}, & \gamma_{\mathrm{sr}} &= \frac{\varepsilon_2 - \varepsilon_3}{1-s_\lambda}, \\ \gamma_{\mathrm{la}} &= \frac{\varepsilon_1 + \varepsilon_2 - 2\varepsilon_3}{3-s_\lambda}, & \gamma_{\mathrm{ra}} &= \frac{2\varepsilon_1 - \varepsilon_2 - \varepsilon_3}{3+s_\lambda}. \end{aligned} \tag{27}$$

The return denominators are

$$\begin{aligned} D_{\mathrm{s}} &= 4\lambda_{\mathrm{L}} s_\lambda^2 + 4\mu(1+s_\lambda^2), & D_{\mathrm{l}} &= 4\lambda_{\mathrm{L}} s_\lambda^2 + \mu(1+s_\lambda)^2 + 2\mu(1-s_\lambda)^2, \\ D_{\mathrm{r}} &= 4\lambda_{\mathrm{L}} s_\lambda^2 + 2\mu(1+s_\lambda)^2 + \mu(1-s_\lambda)^2, & D_{\mathrm{a}} &= 4K s_\lambda^2. \end{aligned} \tag{28}$$

The corresponding branch multipliers are

$$
\begin{aligned}
\Lambda_{\mathrm{s}} &= \frac{f_{\mathrm{tr}}}{D_{\mathrm{s}}}, \\
\Lambda_{\mathrm{l}} &= \frac{\mu((1+s_\lambda)(\varepsilon_1+\varepsilon_2) - 2(1-s_\lambda)\varepsilon_3) + 2\lambda_{\mathrm{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{l}}}, \\
\Lambda_{\mathrm{r}} &= \frac{\mu(2(1+s_\lambda)\varepsilon_1 - (1-s_\lambda)(\varepsilon_2+\varepsilon_3)) + 2\lambda_{\mathrm{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{r}}}, \\
\Lambda_{\mathrm{a}} &= \frac{2K s_\lambda I_1 - \bar{c}_\lambda}{D_{\mathrm{a}}}.
\end{aligned}
\tag{29}
$$

The branch energies are

$$
\begin{aligned}
\Phi_{\mathrm{s}} &= \Phi_{\mathrm{el}} - \frac{1}{2} D_{\mathrm{s}} \Lambda_{\mathrm{s}}^2, \\
\Phi_{\mathrm{l}} &= \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu\left(\varepsilon_3^2 + \frac{1}{2}(\varepsilon_1+\varepsilon_2)^2\right) - \frac{1}{2} D_{\mathrm{l}} \Lambda_{\mathrm{l}}^2, \\
\Phi_{\mathrm{r}} &= \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu\left(\varepsilon_1^2 + \frac{1}{2}(\varepsilon_2+\varepsilon_3)^2\right) - \frac{1}{2} D_{\mathrm{r}} \Lambda_{\mathrm{r}}^2, \\
\Phi_{\mathrm{a}} &= \frac{1}{2} K I_1^2 - \frac{1}{2} D_{\mathrm{a}} \Lambda_{\mathrm{a}}^2.
\end{aligned}
\tag{30}
$$

With these definitions, the scalar potential is the piecewise function

$$
\Phi_{\mathrm{MC,3D}} = \begin{cases}
\Phi_{\mathrm{el}}, & f_{\mathrm{tr}} \le 0, \\
\Phi_{\mathrm{s}}, & \Lambda_{\mathrm{s}} \le \min(\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sr}}), \\
\Phi_{\mathrm{l}}, & \gamma_{\mathrm{sl}} < \gamma_{\mathrm{sr}} \text{ and } \gamma_{\mathrm{sl}} \le \Lambda_{\mathrm{l}} \le \gamma_{\mathrm{la}}, \\
\Phi_{\mathrm{r}}, & \gamma_{\mathrm{sr}} < \gamma_{\mathrm{sl}} \text{ and } \gamma_{\mathrm{sr}} \le \Lambda_{\mathrm{r}} \le \gamma_{\mathrm{ra}}, \\
\Phi_{\mathrm{a}}, & \text{otherwise},
\end{cases}
\tag{31}
$$

The constitutive-AD path differentiates the smooth branches of equation (31) directly with JAX rather than substituting a hand-derived return-map tangent; the branch interfaces remain nonsmooth and are treated as part of the algorithmic constitutive surrogate rather than as a globally smooth continuum potential.

For the strain visualizations in this paper we use the deviatoric norm

$$
\|\varepsilon_{\mathrm{dev}}\| = \sqrt{\varepsilon^T \left(\mathrm{diag}(1,1,1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}) - \frac{1}{3} m m^T\right) \varepsilon}, \qquad m = (1,1,1,0,0,0)^T,
\tag{32}
$$

which matches the engineering-shear convention of equation (22).

Because equations (21) and (31) define an algorithmic endpoint surrogate rather than a path-consistent incremental history functional, the results section reports this family with a separate validation ladder. Direct-branch source-observable agreement for the endpoint surrogate is treated separately from a fixed-λ_{sr} reference-operator comparison. The latter checks source-family behavior under matched geometry, materials, Davis-B reduction parameters, load schedule, and glued-bottom free-DOF mask, but it is not a path-consistent incremental-history validation. The paper therefore keeps the Plasticity3D claims at the level of a problem-specific endpoint surrogate rather than a path-consistent replacement. The related return-mapping, optimization, convex-optimization, and continuation literature [15–18] is used as source-family context, not as evidence that this endpoint surrogate is a full incremental elastoplastic solver.

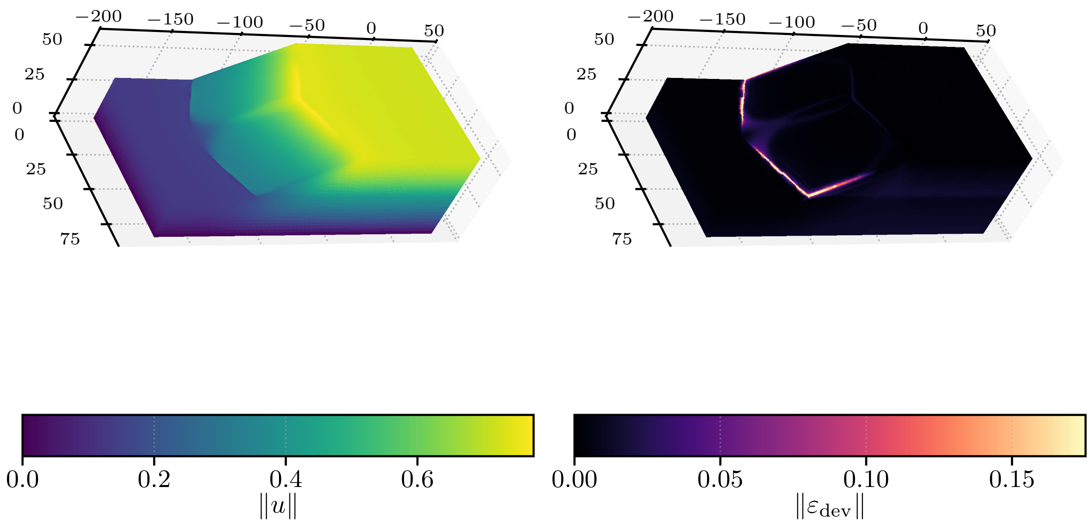


Figure 14: Corrected glued-bottom Plasticity3D showcase at $\lambda_{\mathrm{sr}} = 1.55$: displacement magnitude and surface deviatoric strain for the $P_4(L_2)$ run, both derived from the scalar surrogate in equation (21).

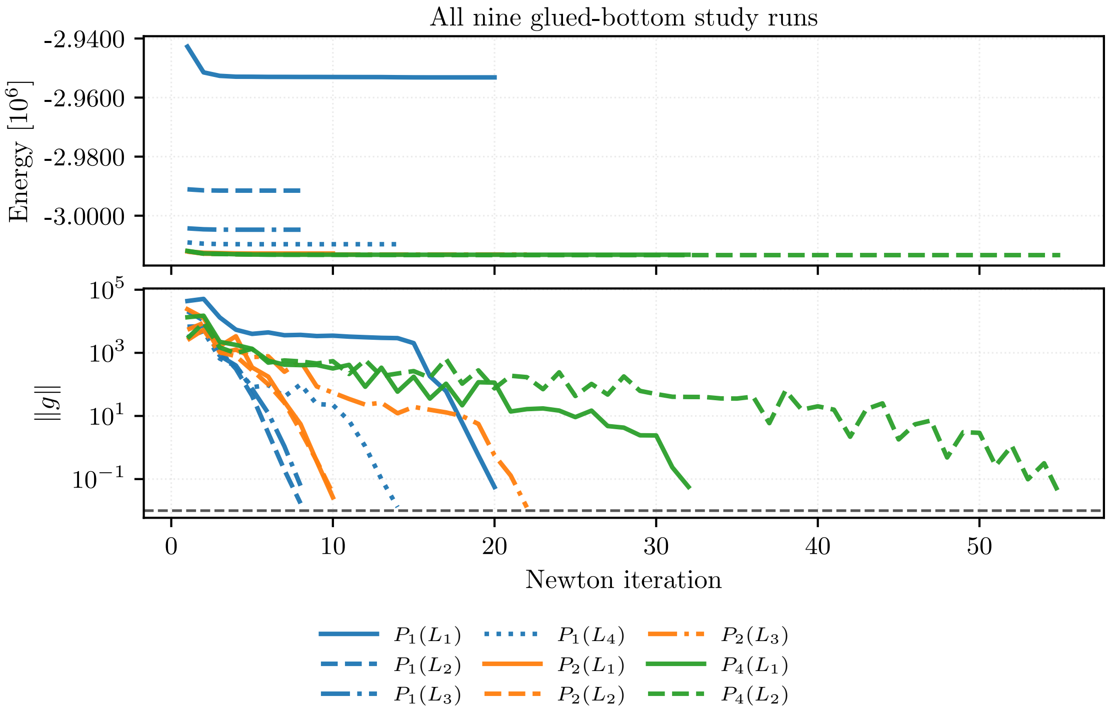


Figure 15: Newton-history convergence for all nine completed corrected glued-bottom Plasticity3D study runs at $\lambda_{\mathrm{sr}} = 1.55$. The upper panel shows the value of the algorithmic surrogate from equation (21); the lower panel shows the gradient norm used as the nonlinear stopping metric, with color indicating degree and line style indicating mesh level.

Element	Free DOFs	Energy	$\|g\|_{\mathrm{final}}$	Wall time [s]
$P_1(L_1)$	10,526	−2,953,167	4.56×10^{-3}	1.79
$P_1(L_2)$	79,024	−2,991,497	1.64×10^{-3}	8.72
$P_1(L_3)$	610,964	−3,004,742	4.07×10^{-3}	17.2
$P_1(L_4)$	4,801,816	−3,009,658	1.16×10^{-3}	96.7
$P_2(L_1)$	79,024	−3,012,813	2.47×10^{-3}	18.3
$P_2(L_2)$	610,964	−3,013,032	2.96×10^{-3}	98.3
$P_2(L_3)$	4,801,816	−3,013,149	1.14×10^{-3}	1330
$P_4(L_1)$	610,964	−3,013,227	5.53×10^{-3}	208
$P_4(L_2)$	4,801,816	−3,013,348	4.42×10^{-3}	2298

Table 9: All nine completed glued-bottom Plasticity3D study rows at $\lambda_{\mathrm{sr}} = 1.55$, reported as endpoint values of equation (21).

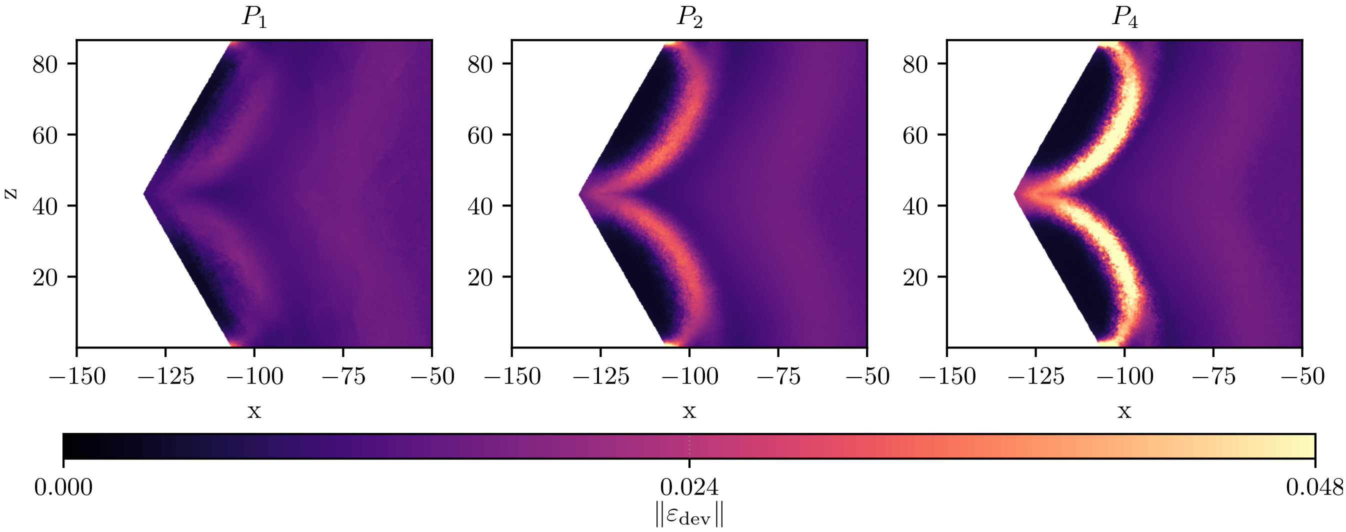


Figure 16: Shared-scale upper y-slice comparison of $\|\varepsilon_{\mathrm{dev}}\|$ from equation (32) at the highest mesh level of each degree line: $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$. The three runs use the corrected glued-bottom solver stack at $\lambda_{\mathrm{sr}} = 1.55$ and are compared on one common color scale.

Plasticity3D is the benchmark where detailed constitutive definitions are most important: the paper later plots values of equation (21) and equation (32), so both quantities are defined explicitly here. Timing, scaling, and degree-versus-resolution evidence for this family are reported in Section 7.7.

5.6 Topology Optimization

The literature model is the standard SIMP-style compliance topology-optimization setting on a cantilever domain, together with filtering and continuation ideas from the classical topology literature [24–27]. In the implemented setting studied here, the topology benchmark is a reduced optimization problem rather than a single equilibrium minimization. The design variable is a latent nodal field $z_h \in Z_h$, mapped to a physical density through the logistic transform

$$\theta_h(z_h) = \theta_{\min} + (1 - \theta_{\min})\sigma(z_h), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{33}$$

For fixed density, the discretization solves for the linear elastic displacement state $u_h(\theta_h)$ solving

$$a_{\theta_h}(u_h, v_h) := \sum_{e=1}^{n_{\mathrm{el}}} |e|\, \theta_e^p \varepsilon(u_h)^T C \varepsilon(v_h) = \ell(v_h) \qquad \forall v_h \in V_{h,0}, \tag{34}$$

on the cantilever domain $\Omega = (0, 2) \times (0, 1)$ with clamped left edge and prescribed traction on a right-edge patch. The reduced objective is

$$\mathcal{J}_h(z_h) = \sum_{e=1}^{n_{\mathrm{el}}} |e| \left[e_e^{\mathrm{frozen}} \theta_e^{-p} + \lambda_V \theta_e + \alpha_{\mathrm{reg}} \left(\frac{\ell_{\mathrm{pf}}}{2} |\nabla \theta_e|^2 + \frac{1}{\ell_{\mathrm{pf}}} W(\theta_e) \right) + \frac{\mu_{\mathrm{move}}}{2} (z_e - z_e^{\mathrm{old}})^2 \right], \tag{35}$$

with $W(\theta) = \theta^2(1 - \theta)^2$, SIMP-style continuation in p, and regularization terms that play a mesh-control role analogous to filtering/regularization in Bourdin [27]. Here e_e^{frozen} denotes the element elastic energy density from the previous mechanics state, λ_V weights the volume term, α_{reg} and ℓ_{pf} define the phase-field-type regularization scale, and μ_{move} weights the latent-field move penalty. These are algorithmic design choices rather than standard SIMP notation. The outer loop alternates mechanics solves for equation (34) with reduced design updates for equation (35).

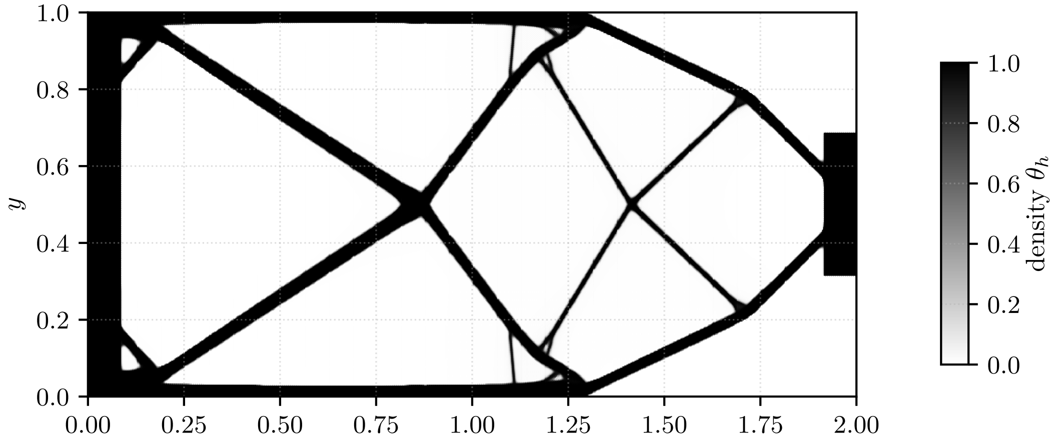


Figure 17: Representative final density field for the parallel topology benchmark.

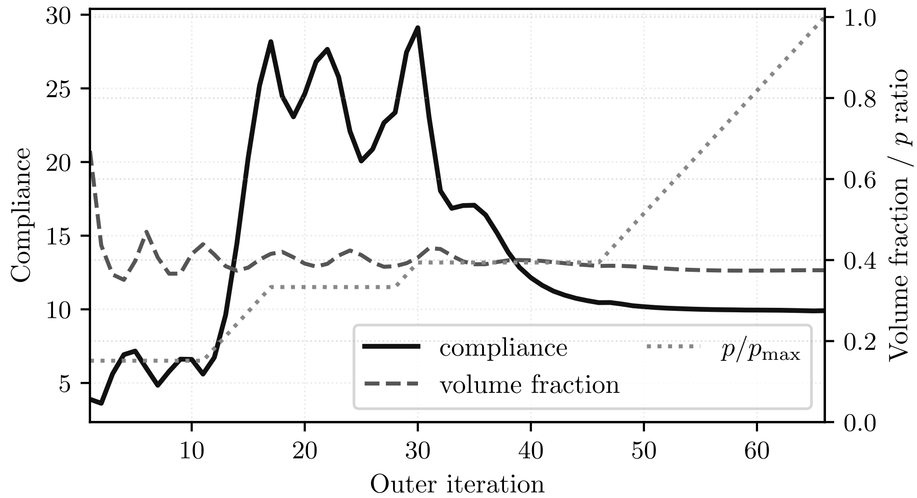


Figure 18: Objective-history view for the reduced topology objective equation (35), with compliance on the left axis and volume fraction plus the continuation schedule in p on the right axis.

Case	Ranks	Outer iters	Compliance	Volume fraction	Wall time [s]
192×96	1	121	4.1557	0.4	10
768×384	32	66	9.9074	0.3749	25.1

Table 10: Representative topology endpoint values from equation (35).

For topology, the relevant notion of "convergence" is the reduced objective history rather than a single equilibrium energy. The reported continuation starts at SIMP exponent $p = 1$, increases p by 0.2 on each outer iteration up to $p = 10$, and treats the fine-grid run as stalled once $p \geq 4$ and both the design change and state change are at most 10^{-6}. Parallel timing and scaling for the same family are reported in Section 7.8.

6 Validation and External Comparisons

This section separates validation and external-comparison evidence from the performance results that follow. The distinction is important because the paper uses different levels of evidence for different claims: Hyperelasticity has a narrow external companion comparison against JAX-FEM, while Plasticity3D has a source-family validation ladder for the implemented endpoint surrogate.

6.1 Comparison Metrics

For vector and field comparisons with nonzero reference norm, we report relative differences of the form

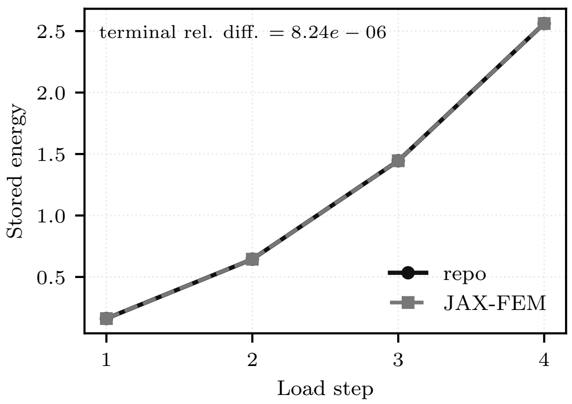
$$\delta_M(a,b) = \frac{\|a-b\|_M}{\|a\|_M}, \tag{36}$$

where a is the reference observable and M denotes the discrete norm used for the comparison. For nodal displacement vectors, M is the Euclidean norm on the compared free degrees of freedom. For sampled profiles and quadrature-derived fields, M denotes the corresponding sampled or quadrature-weighted norm used by the comparison. The reported field and profile observables have nonzero reference norms. Scalar work and energy differences use the same reference-denominator convention; the hyperelastic energy post-comparison uses an absolute denominator floor of 10^{-12}.

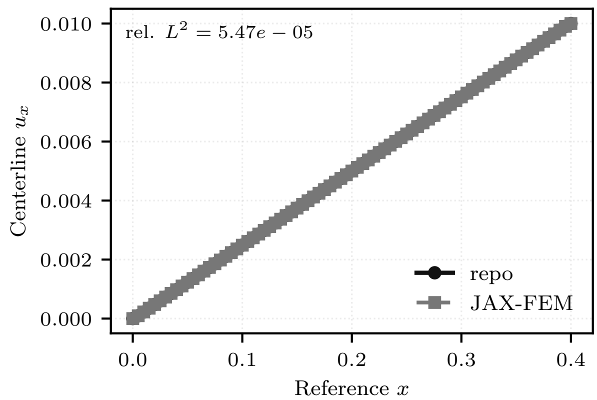
The hyperelastic companion comparison uses a 5% acceptance threshold for the final energy, full-field displacement, centerline displacement, and $u_{\max}$-curve differences. The Plasticity3D Layer 2 acceptance gate uses three thresholds from its validation manifest: at most 0.03 relative difference in the highest successful load factor, at most 0.05 relative L^2 difference for the $u_{\max}(\lambda_{\text{sr}})$ curve, and at most 0.10 relative L^2 difference for the endpoint displacement. The deviatoric-strain and boundary-profile differences in Table 12 are diagnostics, not acceptance gates.

6.2 Hyperelasticity Companion Comparison

The hyperelasticity energy is defined in Section 5.3. The external comparison here is deliberately narrow and uses a serial companion case rather than the 24-step rotating benchmark path. Both implementations use the same level-1 mesh and the same right-face x-translation schedule $d = (0.0025, 0.005, 0.0075, 0.01)$, then post-compare both terminal states with the energy from equation (11). This supports a solution-level implementation comparison for this hyperelastic problem only. It should not be read as a plasticity validation, a broad framework ranking, or constitutive-law identity between the two implementations: the JAX-FEM companion implementation uses a logarithmic-J Piola form, whereas equation (11) uses the $D_1(J-1)^2$ volumetric penalty.



(a) Terminal energy over the matched displacement schedule.



(b) Final centerline u_x profile on the common sample path.

Figure 19: Serial hyperelastic companion comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with energy post-comparison of the terminal states under equation (11).

Group	Quantity	Relative difference	Median wall time [s]	Status
Agreement	final energy	8.24×10^{-6}	–	–
Agreement	full-field displacement relative L^2	2.01×10^{-4}	–	–
Agreement	centerline relative L^2	5.47×10^{-5}	–	–
Agreement	$u_{\max}$ curve relative L^2	8.85×10^{-16}	–	–
Timing	this work serial direct	–	0.176	–
Timing	JAX-FEM UMFPACK serial	–	2.52	–
Contract/gate	same mesh path	–	–	pass
Contract/gate	same displacement schedule	–	–	pass
Contract/gate	agreement thresholds (5%)	–	–	pass
Contract/gate	energy post-comparison	–	–	used

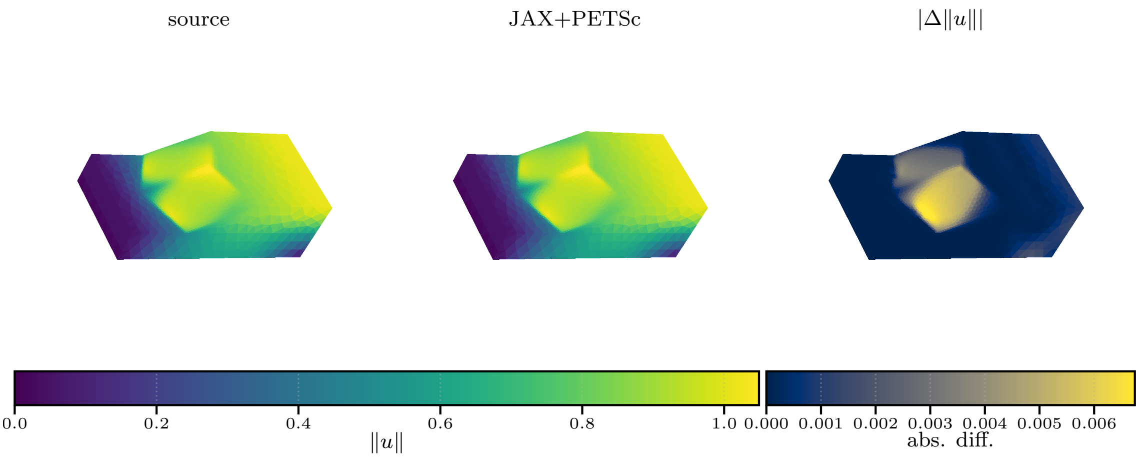
Table 11: Agreement-gated hyperelastic companion comparison against JAX-FEM.

The agreement gate in Table 11 passes on all checked quantities. The final energy relative difference is 8.244×10^{-6}, the full-field displacement relative L^2 difference is 2.011×10^{-4}, the centerline relative L^2 difference is 5.473×10^{-5}, and the $u_{\max}$-curve relative L^2 difference is effectively zero. On this small serial CPU companion case, the JAX+PETSc solve is faster (0.176 s median wall time versus 2.524 s for JAX-FEM), but the scientific role of this table is solution agreement under a narrow hyperelastic comparison contract rather than framework ranking.

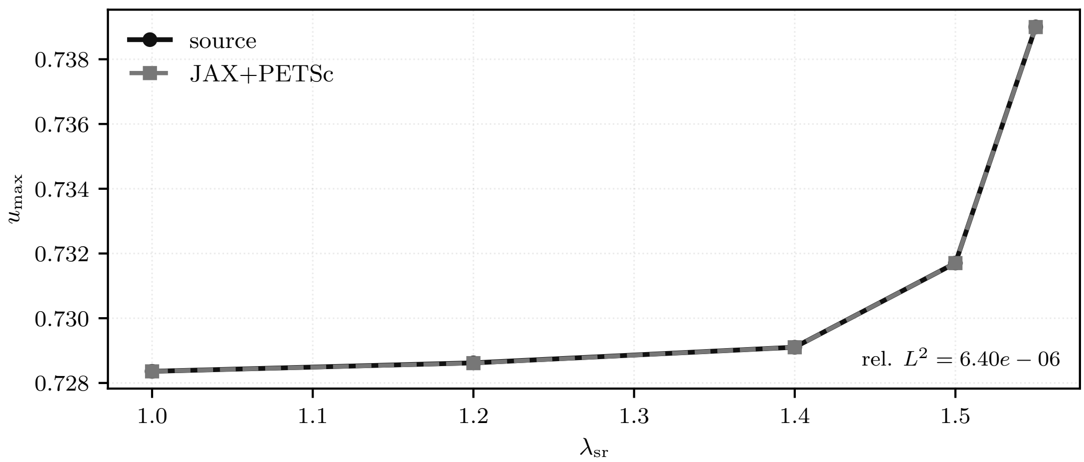
6.3 Plasticity3D Validation Ladder

The Plasticity3D benchmark is tied to a Mohr–Coulomb slope-stability source family, but the object studied here is the algorithmic endpoint surrogate defined in Section 5.5 and equations (21) and (31). The Sysala return-mapping, optimization, convex-optimization, and advanced continuation papers, together with the efficient MATLAB elastoplastic implementation of Čermák et al. [23], provide source-family and method context for this problem class [15–18]; they are not used here as verified numeric paper baselines.

The validation ladder therefore separates two questions. Layer 1A tests direct-branch source-observable agreement for the endpoint surrogate on the $P_2(L_1)$ source-family case. This layer is tight: the work relative difference is 3.877×10^{-5}, the displacement relative L^2 difference is 3.517×10^{-3}, and the deviatoric-strain relative L^2 difference is 8.720×10^{-3}. Layer 2 is different: it is a fixed-λ_{sr} diagnostic against a reference-operator comparison, not proof of path-consistent incremental-history equivalence. After matching the source boundary contract to the glued-bottom free-DOF mask, the validation schedule is capped at the main-figure load level $\lambda_{\mathrm{sr}} = 1.55$. It keeps the highest-successful-λ_{sr} schedule proxy aligned on the tested grid and passes the predefined gates for the $u_{\max}(\lambda_{\mathrm{sr}})$ curve and endpoint displacement: the $u_{\max}(\lambda_{\mathrm{sr}})$ relative L^2 difference is 6.396×10^{-6}, and the endpoint displacement relative L^2 difference is 8.299×10^{-4}. The endpoint deviatoric-strain relative L^2 difference (4.307×10^{-3}) and boundary-profile relative L^2 difference (5.229×10^{-4}) are reported as diagnostics rather than acceptance gates. This comparison uses a matched mesh, material table, Davis-B strength reduction, load schedule, and glued-bottom free-DOF mask, but it remains a fixed-load endpoint check rather than a path-consistent plastic-history reference.



(a) Layer 1A: direct-branch source-observable agreement.



(b) Layer 2: fixed-λ_{sr} reference-operator diagnostic.

Figure 20: Plasticity3D validation ladder. The upper panel documents tight direct-branch source-observable agreement for the endpoint surrogate; the lower panel shows close fixed-λ_{sr} reference-operator agreement after matching the glued-bottom boundary contract.

Layer	Comparison	Relative difference	Status
1A	work	3.88×10^{-5}	–
1A	displacement relative L^2	3.52×10^{-3}	–
1A	deviatoric-strain relative L^2	8.72×10^{-3}	–
2	highest-successful λ_{sr}	0	pass
2	$u_{\max}(\lambda_{\text{sr}})$ relative L^2	6.4×10^{-6}	pass
2	endpoint displacement relative L^2	8.3×10^{-4}	pass
2	endpoint deviatoric-strain relative L^2	4.31×10^{-3}	diagnostic
2	boundary profile relative L^2	5.23×10^{-4}	diagnostic
2	acceptance gate	–	pass

Table 12: Plasticity3D validation summary for direct-branch source-observable agreement (layer 1A) and the fixed-λ_{sr} reference-operator diagnostic (layer 2).

The resulting interpretation is limited but useful. Plasticity3D is supported here by direct-branch source-observable agreement in Layer 1A, while Layer 2 supports the same endpoint surrogate under a fixed-load reference-operator contract. Neither layer is a path-consistent incremental-plasticity history validation.

7 Performance and Solver Behavior

This section turns from problem definition to computational behavior. The benchmark families in Section 5 established the governing equations, discrete energies, and representative states, and Section 6 separated the validation and external-comparison evidence. Here we focus on parallel timings, scaling trends, solver behavior, and the comparative cost of alternative derivative or discretization choices. Evidence from older timing studies is retained when it is still informative for workflow viability, but it is labeled as timing evidence rather than merged into the validation story.

7.1 Globalization Method Comparison

Table 13 isolates the nonlinear globalization choice on four benchmark configurations using the same element assembly path and fixed preconditioner family within each problem. The scalar cases use level-10 meshes obtained by one additional uniform refinement of the level-9 inputs. The p-Laplace row uses 32 MPI ranks with a 30 s wall-time cap. The Ginzburg–Landau row uses 16 MPI ranks with a 120 s cap, because the 32-rank configuration did not reach the nonlinear solve within the fixed setup budget. The hyperelasticity row uses the first eight load steps of the level-4 rank-local assembly path on 32 MPI ranks with a 270 s cap. The Plasticity3D row uses the $P_2(L_1)$ endpoint surrogate at $\lambda_{\text{sr}} = 1.55$ on 32 MPI ranks with a 180 s cap. Timeout rows are retained as results; the table uses solver-reported solve time for completed solves and elapsed wall time for timed-out rows. In the line-search rows the trust-region machinery is disabled; in particular, the hyperelasticity line-search baseline uses GMRES rather than the STCG trust-region subproblem solver. For the Plasticity3D row, all methods are judged against the stricter $\|\nabla \mathcal{F}\|_2 < 10^{-4}$ endpoint gate.

The clean p-Laplace case is the expected easy case: Newton with Armijo line search is sufficient and is the fastest row. Ginzburg–Landau separates the policies more sharply: the same line-search-only policy times out under the fixed cap, while both trust-region variants converge to the same reported energy. For the hyperelasticity excerpt all three policies complete, but Steihaug trust-region globalization without the line-search subproblem safeguard has the largest rejection count and the largest solve time. The hybrid row keeps the trust-region safeguard while reducing both nonlinear iterations and solver time relative to line-search-only Newton. The Plasticity3D endpoint gives the opposite conclusion from the easy scalar case: at the stricter $\|\nabla \mathcal{F}\|_2 < 10^{-4}$ gate, line-search-only Newton reaches the iteration cap near the target, whereas both trust-region policies converge. The hybrid variant is the fastest of the two trust-region rows and also uses fewer Krylov iterations. These rows therefore support a problem-dependent policy choice rather than a universal preference for any single globalization strategy.

Outcome and terminal value						
Benchmark	Method	Ranks	Result	Steps	Time [s]	Energy
p-Laplace L_{10}	Newton + LS	32	completed	1/1	6.95	−0.251,722,449,6
p-Laplace L_{10}	Steihaug TR	32	completed	1/1	11.7	−0.251,722,731,2
p-Laplace L_{10}	Hybrid TR+LS	32	completed	1/1	12	−0.251,722,731,2
Ginzburg–Landau L_{10}	Newton + LS	16	timeout	0/1	121	–
Ginzburg–Landau L_{10}	Steihaug TR	16	completed	1/1	14.3	0.345,624,678,3
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	completed	1/1	14.3	0.345,624,678,3
Hyperelasticity L_4	Newton + LS	32	completed	8/8	194	9.733,145
Hyperelasticity L_4	Steihaug TR	32	completed	8/8	178	9.731,01
Hyperelasticity L_4	Hybrid TR+LS	32	completed	8/8	149	9.730,978
Plasticity3D $P_2(L_1)$	Newton + LS	32	failed	0/1	70.2	−3,012,813
Plasticity3D $P_2(L_1)$	Steihaug TR	32	completed	1/1	20.8	−3,012,813
Plasticity3D $P_2(L_1)$	Hybrid TR+LS	32	completed	1/1	16.4	−3,012,813

Nonlinear and Krylov work						
Benchmark	Method	Ranks	Newton	Krylov	LS evals	TR rejects
p-Laplace L_{10}	Newton + LS	32	15	32	15	0
p-Laplace L_{10}	Steihaug TR	32	29	32	0	0
p-Laplace L_{10}	Hybrid TR+LS	32	29	32	29	0
Ginzburg–Landau L_{10}	Newton + LS	16	0	0	0	0
Ginzburg–Landau L_{10}	Steihaug TR	16	19	25	0	0
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	19	25	19	0
Hyperelasticity L_4	Newton + LS	32	420	6984	2758	0
Hyperelasticity L_4	Steihaug TR	32	314	7245	0	95
Hyperelasticity L_4	Hybrid TR+LS	32	295	6849	499	3
Plasticity3D $P_2(L_1)$	Newton + LS	32	80	858	224	0
Plasticity3D $P_2(L_1)$	Steihaug TR	32	13	96	0	1
Plasticity3D $P_2(L_1)$	Hybrid TR+LS	32	11	73	12	0

Table 13: Local comparison of nonlinear globalization policies. LS denotes Armijo line search and TR denotes Steihaug trust region.

The Ginzburg–Landau row is also rerun as a fixed-budget scalar probe on the level-10 mesh with eight MPI ranks, so every policy reaches the nonlinear solver rather than timing out during setup. Table 14 reports this cleaned scalar-nonconvex comparison separately from the multi-family table above. Its purpose is narrower: it checks whether the trust-region conclusions survive when the setup budget is not the dominant failure mode.

Method	Result	Newton	Krylov	LS evals	TR rejects	Setup [s]	Solve [s]	Energy
Newton + LS	timeout	–	–	–	–	–	–	–
Steihaug TR	completed	19	27	0	0	6.53	28.5	0.345,624,678,3
Hybrid TR+LS	completed	19	27	19	0	6.44	28.5	0.345,624,678,3

Table 14: Ginzburg–Landau level-10 fixed-budget globalization comparison on eight MPI ranks. The rows use the same element assembly and Hypre preconditioner; only the nonlinear globalization policy changes.

7.2 Derivative-Route Comparison

Table 15 isolates the cost of the second-order route on three representative local cases. The p-Laplace rows reuse the curated level-9, 32-rank timing data. The hyperelasticity rows run only the first level-4 load step and intentionally use the replicated element path for both routes, because the rank-local hyperelasticity configuration does not use colored SFD. This makes the row a derivative-route comparison rather than a scaling claim. The Plasticity3D rows use the same $P_2(L_1)$, $\lambda_{\mathrm{sr}} = 1.55$, MUMPS-backed PMG setting as the globalization comparison above.

Across all three benchmark rows, local colored SFD reaches the same reported energy and identical nonlinear/Krylov iteration counts as element AD. The scalar and hyperelasticity rows still show the expected wall-time premium for colored SFD. On the Plasticity3D endpoint, constitutive AD is fastest, while element AD and colored SFD have matching iteration counts and comparable solve times. The colored-SFD Plasticity3D row records 240–420 colors across the 32 ranks; its comparable solve time is therefore explained by batched local HVPs having nearly the same measured

Hessian-callback cost as element AD on this $P_2(L_1)$ case, not by an under-colored finite-difference reconstruction.

Outcome and timing

Benchmark	Route	Result	Time [s]	Energy
p-Laplace L_9	Element AD	completed	1.14	−7.960,003
p-Laplace L_9	Colored SFD	completed	1.45	−7.960,003
Hyperelasticity L_4 step 1	Element AD	completed	11.7	0.151,992,327,5
Hyperelasticity L_4 step 1	Colored SFD	completed	34.6	0.151,992,327,5
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Element AD	completed	16.3	−3,012,813
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Colored SFD	completed	16.4	−3,012,813
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Constitutive AD	completed	4.81	−3,012,813

Work and Hessian construction

Benchmark	Route	Ranks	Newton	Krylov	Hessian [s]	SFD colors
p-Laplace L_9	Element AD	32	6	11	–	–
p-Laplace L_9	Colored SFD	32	6	11	–	–
Hyperelasticity L_4 step 1	Element AD	32	22	492	–	–
Hyperelasticity L_4 step 1	Colored SFD	32	22	492	–	–
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Element AD	32	11	73	6.57	–
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Colored SFD	32	11	73	6.59	240–420
Plasticity3D $P_2(L_1)$ $\lambda_{\mathrm{sr}} = 1.55$	Constitutive AD	32	11	73	1.6	–

Table 15: Local derivative-route comparison. Colored SFD denotes local subgraph finite-difference Hessian recovery; the SFD-color range is shown where the current run records per-rank color counts.

Table 16 adds the degree sensitivity behind the Plasticity3D derivative-route claim. The fixed-work probe uses one Newton linearization on glued-bottom heterogeneous meshes at $\lambda_{\mathrm{sr}} = 1.55$, 32 MPI ranks, and the degree-appropriate PMG hierarchy. This table should be read as derivative cost evidence, not as a convergence-quality replacement for the full Plasticity3D endpoint studies. The same-DOF comparison is the most informative row pair: $P_1(L_2)$ and $P_2(L_1)$ both have 79,024 free unknowns, but the P_2 element stencil is denser. Its colored-SFD route therefore needs 240–420 colors instead of 66–90, raises the Hessian callback from 0.332 to 1.412 s, and raises the rank-maximum RSS from about 0.65 to 1.10 GiB. We omit the P_4 colored-SFD row from this local table; the P_2 same-DOF comparison already exposes the stencil density effect without forcing a large local-memory stress case.

Discretization size

Element	Route	Result	Free DOFs	Elem DOFs	Overlap DOFs	RSS max [GiB]
$P_1(L_1)$	Element AD	fixed work	10,526	12	675	0.5
$P_1(L_1)$	Colored SFD	fixed work	10,526	12	675	0.5
$P_1(L_1)$	Constitutive AD	fixed work	10,526	12	675	0.5
$P_1(L_2)$	Element AD	fixed work	79,024	12	3624	0.6
$P_1(L_2)$	Colored SFD	fixed work	79,024	12	3624	0.6
$P_1(L_2)$	Constitutive AD	fixed work	79,024	12	3624	0.5
$P_2(L_1)$	Element AD	fixed work	79,024	30	3933	0.9
$P_2(L_1)$	Colored SFD	fixed work	79,024	30	3933	1.1
$P_2(L_1)$	Constitutive AD	fixed work	79,024	30	3933	0.6
$P_4(L_1)$	Element AD	fixed work	610,964	105	27177	1.2
$P_4(L_1)$	Constitutive AD	fixed work	610,964	105	27177	1.1

Linearization cost

Element	Route	Free DOFs	Krylov	Solve [s]	Hessian [s]	SFD colors
$P_1(L_1)$	Element AD	10,526	11	0.0806	0.257	–
$P_1(L_1)$	Colored SFD	10,526	11	0.0704	0.256	57–90
$P_1(L_1)$	Constitutive AD	10,526	11	0.0706	0.217	–
$P_1(L_2)$	Element AD	79,024	3	0.232	0.322	–
$P_1(L_2)$	Colored SFD	79,024	3	0.273	0.332	66–90
$P_1(L_2)$	Constitutive AD	79,024	3	0.164	0.257	–
$P_2(L_1)$	Element AD	79,024	5	0.83	1.31	–
$P_2(L_1)$	Colored SFD	79,024	5	0.827	1.41	240–420
$P_2(L_1)$	Constitutive AD	79,024	5	0.304	0.535	–
$P_4(L_1)$	Element AD	610,964	13	8.48	7.04	–
$P_4(L_1)$	Constitutive AD	610,964	13	3.97	3.62	–

Table 16: Plasticity3D one-Newton derivative-route cost versus discretization. Elem DOFs is the local vector element size used by the Hessian callback; overlap DOFs is the maximum local overlap vector size recorded by the assembler.

7.3 p-Laplace

The p-Laplace problem definition and discrete energy are given in Section 5.1 and equation (6). In this clean scalar setting, the numerical story is easiest to interpret: exact element Hessians remain effective in the reported distributed runs, while local-SFD reaches nearly identical discrete energies at higher wall time.

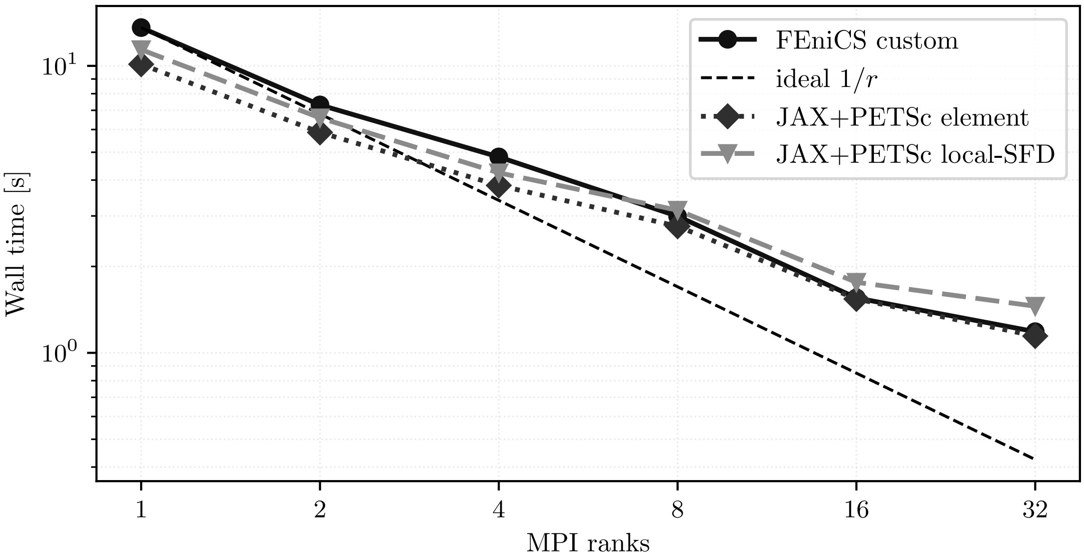


Figure 21: p-Laplace strong scaling on the finest tested grid. All curves correspond to the same discrete problem from equation (6).

7.4 Ginzburg–Landau

The Ginzburg–Landau family from Section 5.2 and equation (10) shows the same high-level derivative story under a more indefinite linearization. The important computational message is not a dramatic separation between exact and approximate second-order routes, but that the distributed solver remains convergent on this nonconvex scalar energy with the same sparse-infrastructure contract.

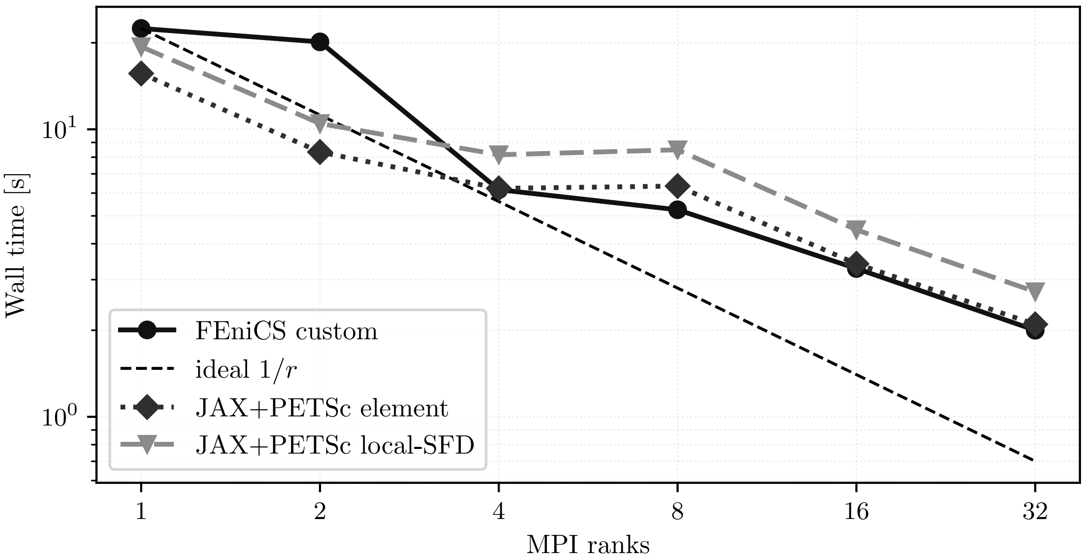


Figure 22: Ginzburg–Landau strong scaling for the tested fine-grid case governed by equation (10).

7.5 Hyperelasticity

The hyperelasticity problem definition, boundary data, and terminal energies are given in Section 5.3 and equation (14). Computationally, this is the benchmark in which globalization matters more than derivative-route choice: the distributed implementation completes the full load path and preserves the terminal energy trend from Section 5.3. The external JAX-FEM companion comparison is reported separately in Section 6.2; the figure below reports only distributed scaling for the load path in equation (14).

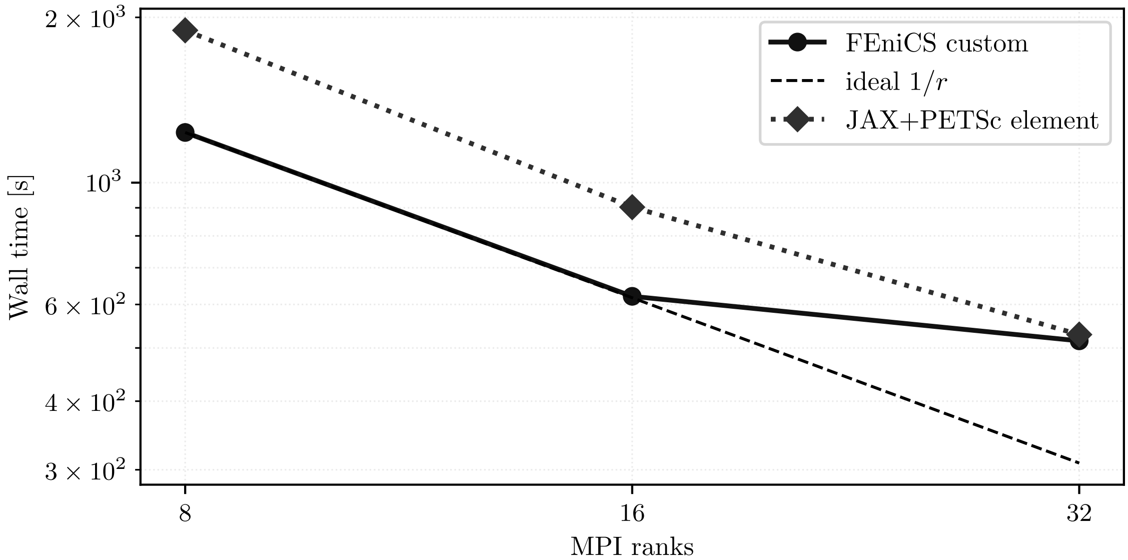


Figure 23: Hyperelasticity strong scaling for the 24-step load path defined by equation (14).

The larger fixed-work multi-node experiment isolates the first level-5 load step after the rank-local hyperelasticity implementation. The CPU cluster nodes used here have two AMD EPYC 7H12 64-core processors and 256 GB RAM [38]; on the tested nodes Slurm exposes the 128 cores as eight NUMA domains with 16 cores each. The runs use one MPI rank per core, contiguous CPU-ID binding, and local memory binding. The mesh is generated procedurally from the uniform beam hierarchy, and the element path assembles from rank-local elements with local PETSc COO preallocation and point-to-point overlap exchange.

The nonlinear solve is the same trust-region Newton path used for the benchmark in section 5.3. The linear subproblems use STCG with a same-hierarchy PMG preconditioner, Chebyshev/Jacobi smoothing, and coarsest level L_2. The coarse solve is PETSc redundant LU with MUMPS: the partial one-node rows use one redundant group over the active ranks, while the multi-node rows use one 128-rank redundant group per node. Figure 24 and table 19 report the application-reported solver total time, not Slurm elapsed time. All rows reach the same first-step energy to the precision shown in Table 19, with 21 Newton iterations and 43 Krylov iterations.

Table 17 records the fixed-work ablation that motivated the rank-local assembly path used in the multi-node runs. The correctness row compares the replicated and rank-local level-4 first-step solve at four MPI ranks; the memory rows use one Newton linearization on the level-5 mesh at 8, 16, and 32 ranks. The table reports both operating-system peak RSS and the assembler's tracked array footprint, making clear whether a row stores global mesh and adjacency data or only local elements plus overlap.

Solve work

Probe	Build	Result	Level	Ranks	Newton	Krylov	Solve [s]
correctness	replicated	completed	L_4	4	26	571	51.2
correctness	rank-local	completed	L_4	4	26	571	50.8
memory	replicated	fixed work	L_5	8	1	1	10.7
memory	rank-local	fixed work	L_5	8	1	1	10.5
memory	replicated	fixed work	L_5	16	1	1	5.57
memory	rank-local	fixed work	L_5	16	1	1	5.5
memory	replicated	fixed work	L_5	32	1	1	3.37
memory	rank-local	fixed work	L_5	32	1	1	3.36

Memory and overlap

Probe	Build	Level	Ranks	RSS max/sum [GiB]	Tracked sum [GiB]	Overlap/owned
correctness	replicated	L_4	4	3.6/14.2	5.49	1.01
correctness	rank-local	L_4	4	3.3/13.2	6.01	1.01
memory	replicated	L_5	8	14.9/118.7	44.34	1.01
memory	rank-local	L_5	8	12.3/98.3	47.87	1.01
memory	replicated	L_5	16	9.4/150.9	45.45	1.03
memory	rank-local	L_5	16	6.3/100.3	47.89	1.03
memory	replicated	L_5	32	6.7/215.6	47.65	1.05
memory	rank-local	L_5	32	3.3/104.9	47.92	1.05

Table 17: Local hyperelasticity distribution and memory ablation. RSS max/sum gives the maximum and summed per-rank resident set size. Tracked sum is the assembler-reported array footprint, and overlap/owned is the summed rank-local overlap DOF count divided by global owned free DOFs.

Table 18 isolates the first-step level-5 PMG choice on the same 32-rank local fixed-work problem. All rows perform the same eight Newton linearizations, so differences in total time are attributable to the linear preconditioner and coarse-solver policy rather than to nonlinear stopping. The energy column is the state after this fixed nonlinear work, so Table 18 is a solver-policy diagnostic rather than an endpoint-accuracy ranking.

Outcome and solver work					
Precond.	Result	Newton	Krylov	Solver [s]	Energy
GAMG	fixed work	8	42	22.3	1.608,441
PMG L_2 + Hypre	fixed work	8	92	23.3	0.525,234,942,7
PMG L_2 + MUMPS	fixed work	8	17	15.3	0.384,891,270,8
PMG L_3 + MUMPS	fixed work	8	17	16.3	0.386,416,603,4
Time breakdown and coarse solve					
Precond.	Coarse	Assembly [s]	PC [s]	Linear [s]	
GAMG	–	9.11	6.41	3.91	
PMG L_2 + Hypre	hypre	10.3	2.93	8.14	
PMG L_2 + MUMPS	mumps, 1 grp.	9.13	2.55	1.91	
PMG L_3 + MUMPS	mumps, 1 grp.	9.14	2.93	2.51	

Table 18: Local hyperelasticity level-5 PMG and coarse-solver sensitivity at 32 MPI ranks with fixed nonlinear work. The PMG rows use the same rank-local element path as the scaling runs.

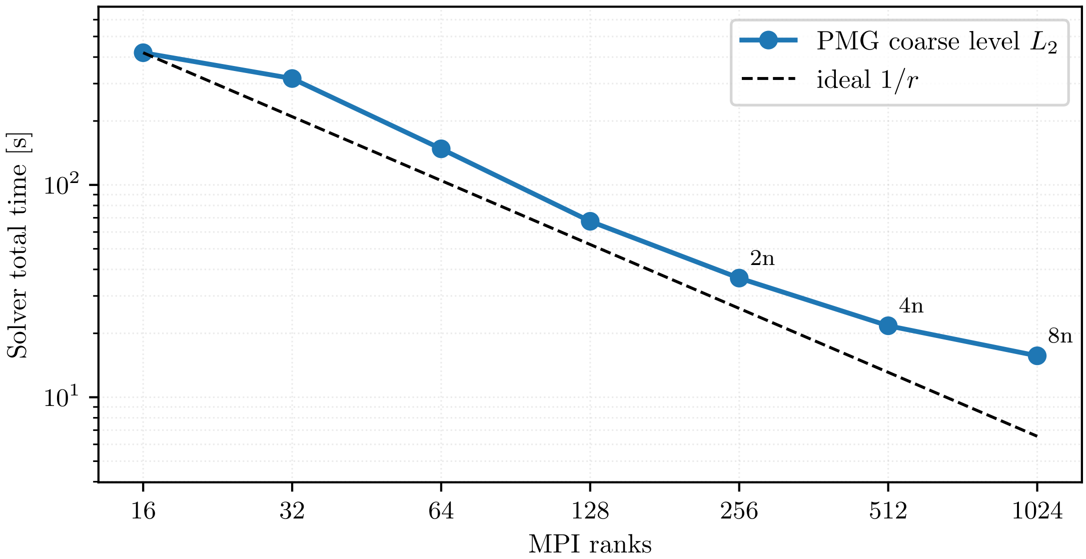


Figure 24: Multi-node CPU level-5 hyperelasticity first-step fixed-work scaling with the PMG coarse level fixed at L_2. Multi-node annotations show node counts.

Ranks	Nodes	Coarse groups	Ranks/group	Solver total [s]	First step [s]	Newton iters	Krylov iters
16	1	1	16	419	367	21	43
32	1	1	32	317	280	21	43
64	1	1	64	148	132	21	43
128	1	1	128	67.4	58.4	21	43
256	2	2	128	36.4	30.7	21	43
512	4	4	128	21.7	17.7	21	43
1024	8	8	128	15.7	12.3	21	43

Table 19: Multi-node CPU level-5 hyperelasticity first-step timings used in figure 24.

7.6 Plasticity2D

The Plasticity2D model and Davis-B reduction are defined in Section 5.4 and equations (17) and (18). For this family, the most informative computational evidence in the present paper is not a single strong-scaling curve. The benchmark section separates the curated endpoint row from the larger fixed-work diagnostics in Figure 13 and table 8, and the source-continuation comparison is collected later in Section 8.

7.7 Plasticity3D

The governing problem, Davis-B reduction, scalar constitutive potential, and deviatoric-strain norm are defined in Section 5.5 and equations (21), (23), (31) and (32). The validation limits for this endpoint surrogate are reported in

Section 6.3. With that interpretation fixed, this section asks two performance questions: which second-order route is fastest on the fixed high-order derivative-route case under matched terminal observables, and how the endpoint surrogate behaves across degree, resolution, and rank count.

Derivative-route ablation. The first question is which second-order route gives the lowest measured cost without changing the reported terminal observables. On the fixed $P_4(L_1)$ case at $\lambda_{\mathrm{sr}} = 1.5$ and 8 MPI ranks, all three routes converge to essentially the same terminal state, with identical Newton and total Krylov counts. This residual-bisection Newton probe uses the Hypre-backed rank-local linear-solver profile, so it should be read as derivative-route timing evidence rather than as the MUMPS-backed PMG profile used by the later $\lambda_{\mathrm{sr}} = 1.55$ convergence studies. The differences are cost differences, not state-quality differences: constitutive AD has the lowest median wall time on this fixed case, with median wall time 222 s and median solve time 191 s, compared with 288 s/255 s for element AD and 372 s/253 s for colored SFD. In Table 20, ω denotes the external-work observable and $u_{\max}$ denotes the maximum displacement magnitude; these columns check that the derivative route does not change the reported terminal observables.

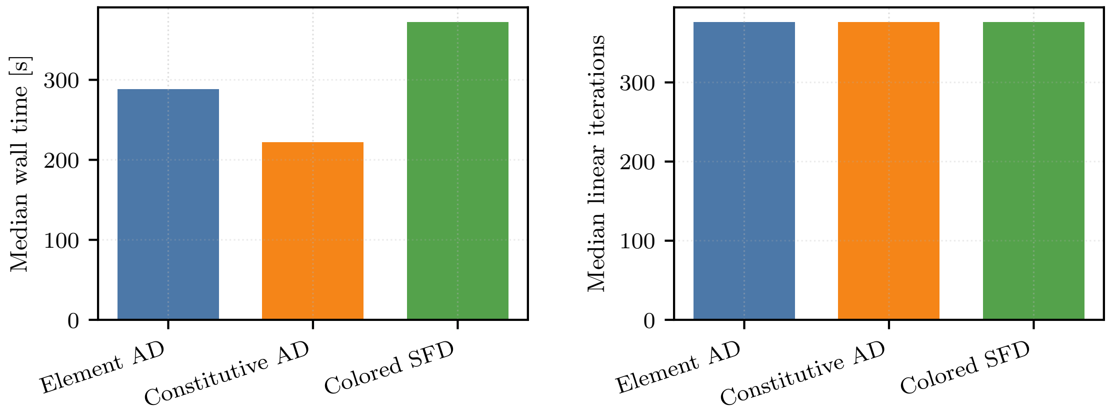


Figure 25: Fixed Plasticity3D derivative-route comparison on $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, and 8 MPI ranks.

Timing and work

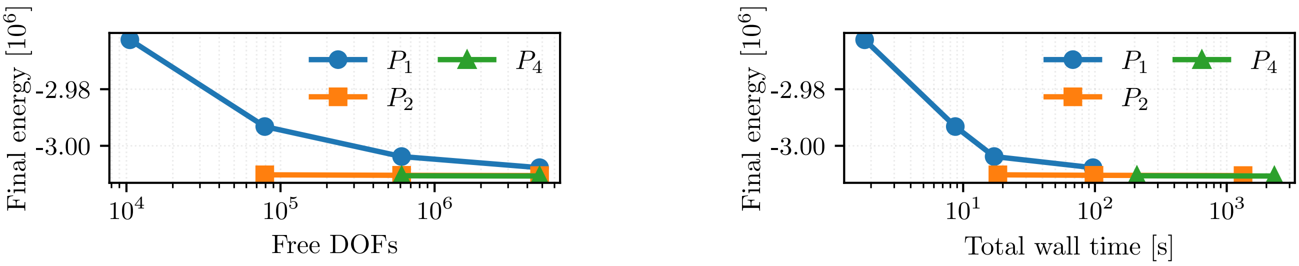
Route	Wall time [s]	Solve time [s]	Newton iters	Krylov iters
Element AD	288	255	8	376
Constitutive AD	222	191	8	376
Colored SFD	372	253	8	376

Terminal observables

Route	Energy	ω	$u_{\max}$
Element AD	−3,010,861	6,027,722	0.731,848
Constitutive AD	−3,010,861	6,027,722	0.731,848
Colored SFD	−3,010,861	6,027,722	0.731,848

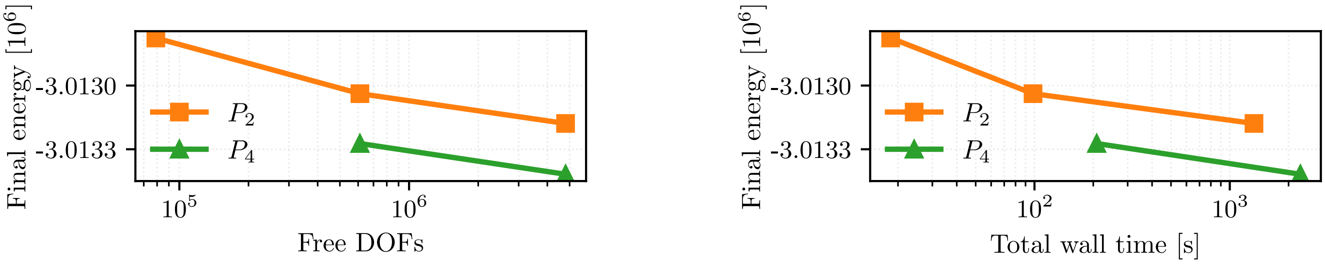
Table 20: Plasticity3D derivative-route comparison on the fixed $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, 8-rank case. These rows use the Hypre-backed rank-local profile and are separate from the MUMPS-backed PMG rows in the $\lambda_{\mathrm{sr}} = 1.55$ studies.

Discretization and scaling. The remaining figures in this subsection answer a different question: how the endpoint surrogate behaves across degree/resolution choices, and how the selected solver configurations scale on the separate $P_4(L_2)$ timing studies.



(a) All degrees: final energy versus free DOFs.

(b) All degrees: final energy versus total wall time.



(c) $P2/P4$ zoom: final energy versus free DOFs.

(d) $P2/P4$ zoom: final energy versus total wall time.

Figure 26: Corrected glued-bottom Plasticity3D degree-versus-resolution study at $\lambda_{\mathrm{sr}} = 1.55$. The top row shows the full $P_1/P_2/P_4$ comparison, while the bottom row zooms to P_2/P_4 so the near-overlap hidden by the coarse P_1 points becomes visible. The left column plots final values of equation (21) against free DOFs; the right column plots the same converged energies against end-to-end wall time. All timings use the same 32-rank configuration.

The zoom row in Figure 26 is essential because the full figure is dominated by coarse P_1 variation. Under the corrected glued-bottom constraint, the first matched comparison is $P_4(L_1)$ versus $P_1(L_3)$ at 610,964 free DOFs. The second matched comparison is $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$ at 4,801,816 free DOFs. The scientific message is therefore that the solver reaches similar terminal energies along both p- and h-refined paths, while higher order on coarser meshes buys lower energy at matched DOFs only by paying a clear wall-time premium.

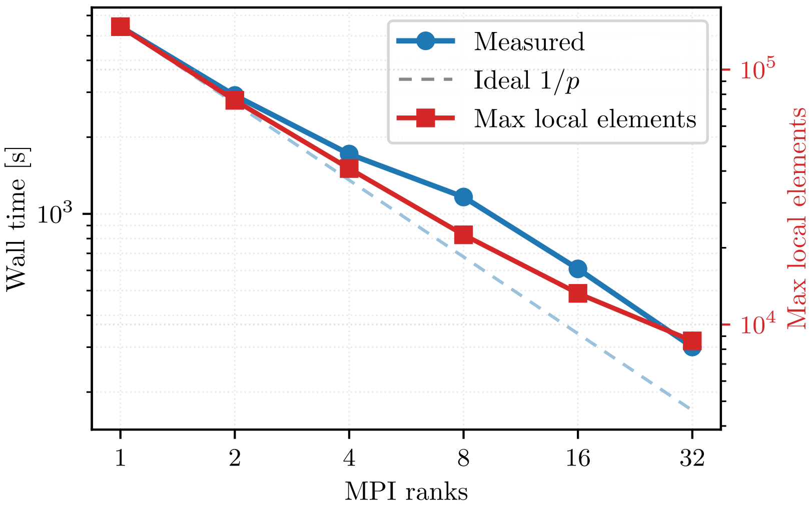


Figure 27: Auxiliary strong scaling for the selected Plasticity3D solver configuration: constitutive-AD local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. This is a separate timing study from the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ results shown in figures 14, 15 and 26.

Ranks	Wall time [s]	Solve time [s]	Speedup	Efficiency	Newton iters	Krylov iters
1	5424	5176	1	1	17	114
2	2914	2777	1.861	0.931	17	114
4	1717	1634	3.16	0.79	17	114
8	1164	1107	4.658	0.582	17	114
16	608	575	8.916	0.557	17	114
32	300	284	18.053	0.564	17	114

Table 21: Auxiliary Plasticity3D strong scaling for constitutive-AD local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. These rows are useful as solver evidence, but they should not be read as a direct timing continuation of the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ benchmark figures.

Figure 27 and table 21 give the paper's largest distributed mechanics timing result for the Hypre-coarse auxiliary PMG profile. This scaling evidence uses the selected $P_4(L_2)$ mesh family but a different load factor, $\lambda_{\mathrm{sr}} = 1.0$, so it complements rather than extends the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ benchmark figures above. Within that timing study, the reported nonlinear and linear iteration counts remain matched on 1/2/4/8/16/32 ranks, while the end-to-end time drops from 5424 s on one rank to 300 s on 32 ranks. Additional reference-formula assembly and continuation evidence is reported later in Section 8.

The converged glued-bottom $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ scaling run repeats the benchmark with the constitutive-AD rank-local assembly and MUMPS-backed PMG profile. The PMG hierarchy is the same-mesh $P_4 \to P_2 \to P_1$ hierarchy, the nonlinear method is Armijo Newton, and the PETSc relative KSP residual tolerance is 1×10^{-2}. The nonlinear stop requires both relative correction below 0.002 and gradient norm below one percent of the first Newton gradient. Figure 28 and table 22 compare the single-node CPU sweep with the multi-node CPU sweep at 16 MPI ranks per node. All plotted rows converge in 29 Newton iterations and 1240 total Krylov iterations to the same energy and gradient norm to the precision shown in the table. The apparently large final gradient norm is below the relative gradient target; the table therefore reports both $\|g\|$ and the ratio $\|g\|/\|g\|_{\mathrm{target}}$. The timing comparison is not confounded by different nonlinear work.

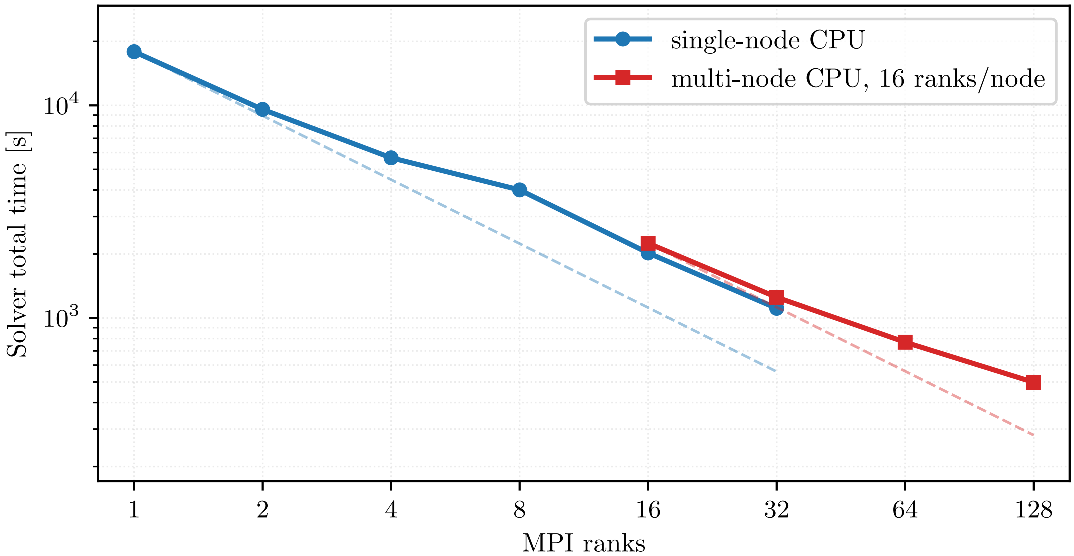


Figure 28: Single-node and multi-node CPU converged scaling for the glued-bottom Plasticity3D $P_4(L_2)$ case at $\lambda_{\mathrm{sr}} = 1.55$. Dashed references are ideal $1/r$ lines anchored separately to the first point of each platform.

Platform	Ranks	Nodes	Solver total [s]	Solve [s]	Newton iters	Krylov iters	Energy	$\|g\|$	$\|g\|$/target
single-node CPU	1	–	17,890	17,615	29	1240	−3,013,348	10.594	0.525
single-node CPU	2	–	9574	9412	29	1240	−3,013,348	10.594	0.525
single-node CPU	4	–	5667	5564	29	1240	−3,013,348	10.594	0.525
single-node CPU	8	–	4003	3947	29	1240	−3,013,348	10.594	0.525
single-node CPU	16	–	2022	1993	29	1240	−3,013,348	10.594	0.525
single-node CPU	32	–	1110	1093	29	1240	−3,013,348	10.594	0.525
multi-node CPU	16	1	2246	2204	29	1240	−3,013,348	10.594	0.525
multi-node CPU	32	2	1251	1227	29	1240	−3,013,348	10.594	0.525
multi-node CPU	64	4	769	754	29	1240	−3,013,348	10.594	0.525
multi-node CPU	128	8	498	488	29	1240	−3,013,348	10.594	0.525

Table 22: Converged timings for the Plasticity3D $P_4(L_2)$, $\lambda_{\text{sr}} = 1.55$ run in figure 28. Single-node rows use one CPU node; multi-node rows use 16 ranks per CPU node. The nonlinear stop requires relative correction below 0.002 and final gradient below one percent of the first Newton gradient.

The distributed layout diagnostics in Table 23 explain why the observed scaling cannot follow the ideal reference indefinitely. The maximum local overlap vector shrinks with rank count, but the sum of rank-local overlap DOFs increases from about the owned problem size on one rank to more than four times the owned size on 128 ranks. This overlap replication comes from subdomain boundaries and high-order element support, so it is a decomposition cost rather than duplicated global mesh storage.

Platform	Ranks	Nodes	Max local DOFs	Overlap / owned DOFs
single-node CPU	1	–	4,859,595	1.01
single-node CPU	2	–	2,495,265	1.05
single-node CPU	4	–	1,284,351	1.11
single-node CPU	8	–	669,783	1.21
single-node CPU	16	–	351,219	1.41
single-node CPU	32	–	182,613	1.83
multi-node CPU	16	1	351,219	1.41
multi-node CPU	32	2	182,613	1.83
multi-node CPU	64	4	91,821	2.66
multi-node CPU	128	8	50,637	4.31

Table 23: Distributed layout diagnostics for the converged Plasticity3D scaling run. "Max local DOFs" is the largest local-overlap vector size on any rank; "Overlap / owned DOFs" is the summed rank-local overlap DOF count divided by the global owned free-DOF count.

7.8 Topology Optimization

The reduced design problem, mechanics subproblem, and objective history are defined in Section 5.6 and equations (34) and (35). The numerical question here is distributed outer-loop stability: the mechanics solve and the design update must remain synchronized under continuation in p and move-regularized updates. Because the outer-loop work can vary with the continuation schedule, the timing evidence here should be read as end-to-end workflow scaling rather than fixed-work strong scaling.

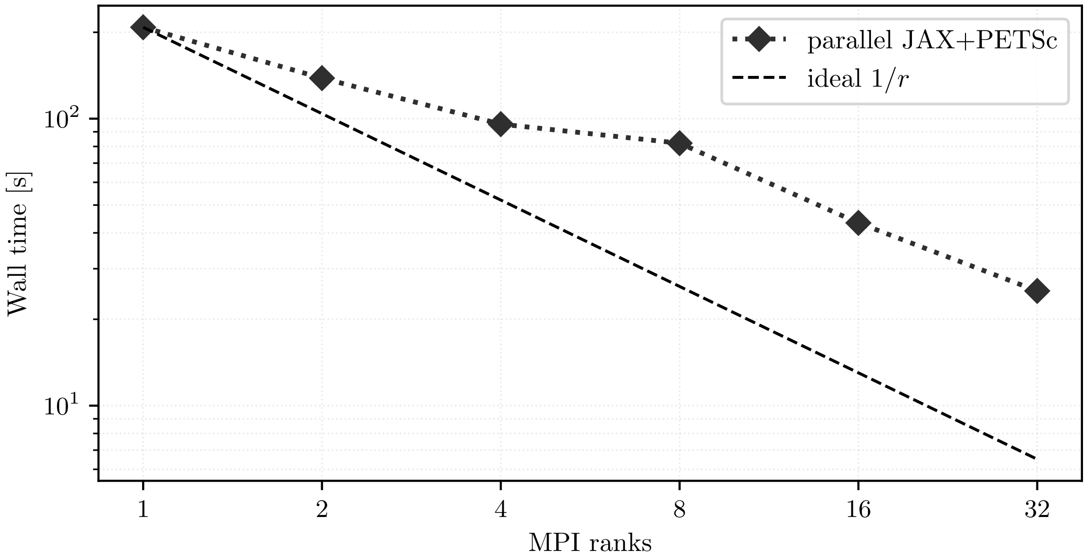


Figure 29: Topology end-to-end parallel timing trend for the fine-grid run. The plotted wall times correspond to the coupled reduced-design workflow defined by equation (35).

Ranks	Wall time [s]	Solve time [s]	Outer iters	p	Compliance	Volume fraction
1	208	200	65	4.2	9.1559	0.3884
2	138	134	72	5.6	8.9473	0.3932
4	95.7	92.8	69	4.6	9.1684	0.385
8	82.1	80.1	67	7	9.2178	0.3932
16	43.3	41.9	66	5.8	9.6859	0.3798
32	25.1	23.7	66	6.6	9.9074	0.3749

Table 24: Parallel topology summary for the fine-grid study, reported for the reduced objective in equation (35).

Table 25 complements the end-to-end scaling table with a fixed-schedule rank-consistency probe on a smaller 384×192 grid. The run disables early convergence and stall stopping, uses the same initial design and smoother continuation schedule ($p \leftarrow p + 0.1$) on all ranks, and compares the saved final density field against the one-rank reference after 40 outer iterations. All rows complete the fixed work. The compliance difference stays below 0.2%, and the density relative L^2 difference remains below 0.04, which checks that the distributed design-gradient exchange and mechanics coupling are no longer visibly rank-dependent on this diagnostic grid.

Ranks	Result	Outer	Solve [s]	Compliance	Volume	p	$\Delta C/C_1$	Density rel. L^2
1	fixed work	40	48.3	15.56	0.3085	5	0	0
2	fixed work	40	28.5	15.56	0.3083	5	1.29×10^{-6}	1.37×10^{-2}
4	fixed work	40	19.7	15.5308	0.3086	5	1.87×10^{-3}	1.91×10^{-2}
8	fixed work	40	16	15.5665	0.3081	5	4.16×10^{-4}	3.68×10^{-2}
16	fixed work	40	8.23	15.5606	0.3084	5	4.15×10^{-5}	1.47×10^{-2}
32	fixed work	40	6.43	15.5637	0.3083	5	2.41×10^{-4}	9.71×10^{-3}

Table 25: Topology fixed-schedule rank consistency. Each row completes the same 40 outer iterations; compliance and density differences are relative to the one-rank row.

The topology benchmark completes the paper's computational picture. Here the mainline JAX+PETSc path is less about exact Hessians than about stable distributed design-mechanics coupling, while pure JAX remains the compact serial reference path discussed earlier in Table 4.

7.9 Synthesis of the Computational Evidence

The results above should be read as a coupled solver study rather than as a collection of isolated benchmark timings. First, the globalization tables show that the nonlinear policy is problem-dependent: line-search Newton is adequate on the clean p-Laplace case, while trust-region safeguards are beneficial on the nonconvex Ginzburg–Landau probes and necessary for the stricter $P_2(L_1)$ Plasticity3D globalization probe. Second, the derivative-route comparisons show that access to several second-order routes is useful in practice. Colored SFD is a credible verification and fallback route on the tested fixed-size problems, while constitutive AD gives the lowest measured wall time on the fixed high-order

Plasticity3D derivative-route case. Third, the hyperelasticity distribution and PMG ablations, together with the Plasticity3D partitioning diagnostics, show that sparse ownership, overlap size, coarse solves, and preconditioner policy are part of the numerical method, not merely software detail. Finally, the hyperelasticity, Plasticity3D, and topology scaling runs show that the same JAX+PETSc mainline remains operational when the problem is large enough that serial reference formulations mainly serve validation, debugging, or comparison roles.

This synthesis is also the boundary of the paper's claim. The experiments support a nonlinear FEM toolset in which local differentiable models, distributed sparse assembly, nonlinear globalization, Krylov linear algebra, and problem-specific preconditioners are developed and assessed together. They do not imply that one backend or one globalization policy dominates all finite-element workflows, nor do they turn the Plasticity3D endpoint surrogate into a path-consistent incremental-history validation.

8 Additional Solver Evidence

This section collects supporting solver evidence that is methodologically useful but too narrow for the main benchmark narratives.

8.1 Plasticity3D Reference-Formula and Fixed-Operator PMG Comparisons

The primary Plasticity3D numerical story in Section 7.7 is the rank-local constitutive-AD path. The tables below add two solver-policy comparisons on the same discrete problem from equation (21): a matched reference-formula assembly comparator and a fixed reference-operator PMG variant.

Ranks	Constitutive wall [s]	Reference wall [s]	Constitutive solve [s]	Reference solve [s]	Ratio
4	1717	951	1634	890	1.805
8	1164	697	1107	656	1.671
16	608	374	575	350	1.628
32	300	277	284	260	1.085

Table 26: Matched constitutive-assembly versus reference-formula assembly comparison for the same PMG solver stack. At each listed rank both rows converge in the same number of nonlinear and linear iterations, so this is a timing-only comparison on equation (21).

Outcome and wall time

Ranks	Route	Status	Wall time [s]
4	constitutive-AD PMG solver	completed	2116
4	reference-formula assembly PMG variant	completed	988
8	constitutive-AD PMG solver	completed	1874
8	reference-formula assembly PMG variant	completed	898
16	constitutive-AD PMG solver	failed	1304
16	reference-formula assembly PMG variant	failed	601
32	constitutive-AD PMG solver	completed	348
32	reference-formula assembly PMG variant	failed	398

Newton–Krylov work

Ranks	Route	Newton iters	Krylov iters	Final gradient norm
4	constitutive-AD PMG solver	24	204	6.53×10^{-3}
4	reference-formula assembly PMG variant	21	210	2.49×10^{-3}
8	constitutive-AD PMG solver	33	238	2.53×10^{-3}
8	reference-formula assembly PMG variant	27	246	9.53×10^{-3}
16	constitutive-AD PMG solver	50	196	2.21×10^{2}
16	reference-formula assembly PMG variant	50	177	7.35×10^{2}
32	constitutive-AD PMG solver	29	250	2.24×10^{-3}
32	reference-formula assembly PMG variant	50	193	8.25×10^{1}

Table 27: Fixed reference-operator PMG alternative on the same Plasticity3D problem. The status column distinguishes completed rows from rank counts that hit the Newton cap before reaching the target gradient tolerance.

These tables support two narrower statements. First, reference-formula assembly has lower reported wall time than rank-local assembly when both are embedded in the same PMG solver stack, although the gap narrows at higher rank counts. Second, the fixed reference-operator PMG profile is not the default used in the present study: it may appear favorable on some rows, but it does more nonlinear and Krylov work and converges less consistently across the reported runs. The rank-local constitutive-AD path remains the mainline because it uses the same local differentiable formulation as the derivative-route and converged scaling studies; the reference-formula rows are retained to expose solver-policy sensitivity on matched discrete problems.

8.2 Plasticity2D Source Continuation Evidence

For Plasticity2D, the source-continuation comparison is not the mainline method of the paper, but it is still useful because it isolates how much preconditioner policy matters even when the outer continuation logic is held fixed.

Policy	Ranks	Runtime [s]	Init Krylov iters	Continuation Krylov iters	Final λ_{sr}
fixed PMG-shell smoother	8	489	58	733	1.568,403
fixed PMG-shell smoother	32	198	60	829	1.568,334

Table 28: Source continuation with the fixed PMG-shell smoother policy at 8 and 32 ranks. This is supporting solver evidence only; it is not part of the paper's mainline solver story.

8.3 Code, data, and reproducibility availability

The implementation discussed in this paper is available at `https://github.com/Beremi/fenics_nonlinear_energies`. The submission archive should include the manuscript-generation scripts, generated table and figure assets, literature-source audit, raw benchmark summaries, and asset validation checks used to build the reported PDF. A citable source or artifact archive DOI should be supplied in the final submission metadata if required by the target journal; this journal-neutral draft does not assert one.

9 Discussion

The results support four main conclusions.

Constitutive AD is fastest in the fixed high-order Plasticity3D derivative test. Figure 25 and table 20 show a clean cost-quality picture on the fixed $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, 8-rank case. All three routes reach essentially the same terminal state with the same nonlinear and Krylov counts, so the comparison is about cost rather than about different converged solutions. In this fixed test, constitutive AD preserves the scalar-energy implementation contract while reducing wall time substantially relative to full element AD and colored SFD.

Validation has to be separated by claim level. The Plasticity3D validation ladder in Figure 20 and table 12 supports a scoped mechanics interpretation. The tested endpoint surrogate agrees closely with the source-family observables reported here, and the fixed-λ_{sr} reference-operator comparison satisfies the endpoint field-agreement thresholds after matching the glued-bottom boundary contract. The evidence therefore supports the intended endpoint surrogate for the tested cases, but it does not establish path-consistent incremental elastoplastic-history equivalence or a verified numeric reproduction of the Sysala source-family papers.

Matched external baselines are useful when the contract is narrow. The hyperelastic companion comparison against JAX-FEM in Figure 19 and table 11 is informative because the mesh, prescribed displacement schedule, and terminal-state energy evaluation are explicitly matched. The terminal states are post-compared with the energy in equation (11); the resulting energy and field differences are very small. The comparison is useful precisely because it is narrow and hyperelastic-only. The paper should not generalize it into a broad ranking between frameworks or into plasticity validation, but it does strengthen the manuscript by showing that at least one recent external implementation can be matched on a shared hyperelastic problem contract.

Scope of the evidence. The same evidence also defines the scope of the conclusions. The auxiliary Plasticity3D strong scaling study at $\lambda_{\mathrm{sr}} = 1.0$ remains timing evidence, while the newer glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ scaling sweep is the converged comparison tied to the benchmark. The topology study is reported as end-to-end workflow scaling rather than fixed-work strong scaling. Some reference-operator PMG studies remain solver-engineering diagnostics rather than fully symmetric scientific baselines. These boundaries give the proper interpretation of the contribution: a software-and-methods workflow with explicit derivative-route choices, explicit validation surfaces, and explicit agreement limits.

The broader methodological lesson is that differentiable FEM becomes most scientifically useful when local differentiation and sparse solver engineering are designed together. Automatic differentiation alone does not settle the behavior of large nonlinear FEM workflows; globalization, sparse ownership, overlap replication, coarse-level policy, preconditioning, and the interpretation of surrogate models matter just as much. Within that scoped claim, the toolset contributes a documented nonlinear FEM workflow rather than a universal differentiable-PDE platform or a new constitutive theory.

10 Conclusion

This paper presented a computational toolset for nonlinear finite-element solves and optimization with a mainline JAX+PETSc distributed path. The central contribution is not a new constitutive theory or a universal differentiable-PDE abstraction, but an implementation pattern in which local JAX differentiation, sparse PETSc nonlinear solves, and multiple second-order routes are kept together across a heterogeneous benchmark suite.

The Plasticity3D evidence is intentionally narrower than a broad mechanics claim. For the tested cases, the endpoint surrogate agrees closely with the reported source-family observables and passes the predefined acceptance quantities, while the evidence remains below the level needed for path-consistent incremental-history validation. Constitutive AD gives the lowest measured wall time in the fixed high-order derivative-route study. On hyperelasticity, the companion comparison against JAX-FEM shows that the JAX+PETSc solver can agree very closely with a recent external implementation when the mesh and prescribed displacement schedule are matched and terminal states are post-compared with equation (11). The distributed Plasticity3D and topology runs then show that the same toolset extends beyond the small validation cases to larger sparse nonlinear solves.

Future work should follow the boundaries made visible by these results:

- extend the validation surface for path-dependent mechanics beyond the present surrogate-versus-reference-operator comparison, including a true incremental-history reference;
- add more agreement-focused external baselines on tightly matched problem contracts;
- continue reducing linear-setup and multigrid costs in the largest 3D mechanics workflows.

The next scientific step is to extend the same validation discipline to additional nonlinear multiphysics and design problems.

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